Absent Multi-view Semi-supervised Classification

Wenzhang Zhuge, Tingjin Luo, Hong Tao, Chenping Hou*, *Member, IEEE*, Dongyun Yi*

***Abstract*—With the advent of vast data collection ways, data are often with multiple modalities or coming from multiple sources. Traditional multi-view learning often assumes that each example of data appears in all views. However, this assumption is too strict in some real applications such as multi-sensor surveillance system, where every view suffers from some data absent. In this paper, we focus on how to classify such incomplete multi-view data in semi-supervised scenario and a method named Absent Multi-view Semi-supervised Classification (AMSC) has been proposed. Specifically, partial graph matrices are constructed independently by anchor strategy to measure the relationships among between each pair of present samples on each view. And to obtain unambiguous classification results for all unlabeled data points, AMSC learns view-specific label matrices and a common label matrix simultaneously. AMSC measures the similarity between pair of view-specific label vectors on each view by partial graph matrices, and consider the similarity between view-specific label vectors and class indicator vectors based on the common label matrix. To characterize the contributions of different views, the p-th root integration strategy is adopted to incorporate the losses of different views. By further analyzing the relation between the p-th root integration strategy and exponential decay integration strategy, we develop an efficient algorithm with proved convergence to solve the proposed non-convex problem. To validate the effectiveness of AMSC, comparisons are made with some benchmark methods on real-world datasets and in the document classification scenario as well. The experimental results demonstrate the advantages of our proposed approach.**

***Index Terms*—Multi-View Learning, Incomplete Data, Semi-supervised Classification, Anchor Strategy**

I. Introduction

With the advent of vast data collection ways, in many applications such as image categorization and cross-language document classification, data are with multiple modalities or coming from multiple sources. The collected unlabeled data is usually sufficient but labeled data is often scarce because labeling data is still a time consuming as well as biased task [1], [2], [3]. Multi-View Semi-Supervised Learning (MVSSL) provides a natural way to deal with such data, where each view corresponds to one modality or information channel. By enforcing view-specific predictor consensus on the unlabeled data, MVSSL takes advantage of large amounts of unlabeled examples and multiple views of data. MVSSL has been shown to be efficient in many important tasks including classification [4], [5], [6], and dimensionality reduction [7], [8], [9]. In particular, we focus on the classification problem.

Although existing multi-view semi-supervised classification methods [3], [10], [11], [12], [13], [14], [15] generate promising results, most of them assume multiple views are complete, i.e., each example of data appears in all views. However, in real-world applications, it is often the case that every view suffers from some data absent, which results in partial multi-view data. For example, in multi-sensor surveillance system, multiple cameras capture the same scene from different angles at the same time. However, some sensors could be temporarily out of action for natural or man-made reasons, thus we will obtain some multi-view samples with some missing views. Another example is web image retrieval. Not all web images are associated with text descriptions and the image itself may be inaccessible due to deletion or invalid url. Such incompleteness makes it impracticable to apply traditional multi-view methods on these data directly. Therefore, how to efficiently classify partial multi-view data in semi-supervised learning becomes a practical and important problem.

A natural thought is to delete the incomplete data, then apply existing multi-view methods on the rest data. However, this thought is clearly not suitable since the purpose is to classify all unlabeled samples to their corresponding classes and it causes the waste of data. To make full use of available data, another thought is to apply a single-view semi-classification method on each view separately and independently to obtain the classification results of their present samples, then determine the label of an unlabeled data according to the vote of the classification results on its present views. The main limitation of this thought is that it fails to make full use of the information among multiple views, and the the performance may suffer when there are unreliable views.

To obtain merits from multi-view learning, there are mainly two feasible strategies: The first strategy is to utilize one of completion methods to fill the missing entries, and then apply a traditional multi-view semi-supervised method for classification. Most completion methods [16], [17] focus on predicting the missing entries of the incomplete data matrix, and several recent works [18], [19] are designed for incomplete multiple kernels. The main limitation of this strategy is that the completion procedure introduces uncertain information, which may lead to a deviation especially when incomplete rate is large. Besides, as a preprocessing, the completion procedure increases the computation complexity. The second strategy is to use one of incomplete multi-view learning methods such as [20], [21], [22], [23], [24], [25], [26], [27], [28], [29] to learn a common representation matrix, and then apply a single-view semi-supervised classification method on it. The

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Wenzhang Zhuge, Tingjin Luo, Hong Tao and Chenping Hou are with College of Liberal Arts and Science, National University of Defense Technology, Changsha 410073, China. E-mail: zgwznudt@yeah.net, tingjinluo@gmail.com, taohong.nudt@hotmail.com, hcpnudt@hotmail.com.

Dongyun Yi is with School of Mathematics and Computing Science, Hunan First Normal University, Changsha 410205, China and also with College of Liberal Arts and Science, National University of Defense Technology, Changsha 410073, China. E-mail: dongyun.yi@gmail.com.

main limitation of this strategy is that it disconnects processes of representation learning and semi-supervised classification, which prevents the two learning processes from negotiating with each other to achieve the optimal performance. The method [30] connect these two processes and achieve convincing performance. However, it cannot well utilized the local information of data and has high computational complexity, which limits its effectiveness of efficiency.

In this paper, to effectively and efficiently classify the incomplete multi-view data in semi-supervised scenario, we propose a new graph-based method, namely Absent Multi-view Semi-supervised Classification (AMSC). Since every view may suffer some samples missing, to utilize the complementary information of multiple views, AMSC first adopts anchor strategy to construct partial similarity matrices on each view separately, which measure relationships between each pair of present samples on each view. To obtain unambiguous classification results from multiple views, AMSC learns view-specific label matrices and a common probability label matrix simultaneously. To establish a reasonable label propagation mechanism, AMSC defines both intra-view similarity loss and extra-view similarity loss. Moreover, to distinguish the importance of multiple views, AMSC incorporates the losses of different views by the p-th root integration strategy [31]. In addition, we develop an efficient algorithm to solve the non-convex problem. The contributions of this paper are summarized as follows.

- We introduce a graph-based semi-supervised classification model AMSC designed for incomplete multi-view data, which classifies unlabeled data without introducing uncertain information. To develop the AMSC model, we define the extra-view similarity loss together with the intra-view similarity loss, which enables the view-specific label propagation to better fulfill the ultimate goal, i.e., clear classification. Besides, an instance weight learning mechanism is established accordingly, which assigns suitable weights for each instance according to their classification certainty automatically.
- We develop an efficient algorithm with proved convergence to solve the resultant non-convex problem of AMSC which has complex constraints. By adopting anchor-strategy to construct partial similarity matrix, the developed algorithm has linear complexity w.r.t. to both data size and dimensionality.
- We analyze the relation between the p-th root integration strategy and the exponential decay incorporation strategy, which extends the unified weight learning paradigm [31] for multi-view learning.
- The effectiveness of the proposed algorithm is verified by systematical experimental results on nine datasets and in the document categorization application.

The rest of paper is organized as follows. Section II reviews the related works. The details of our proposed AMSC are introduced in Section III and IV, respectively. Experimental results on nine benchmark datasets are displayed in Section V. AMSC is applied into document classification in Section VI. Finally, we conclude this paper in Section VII.

II. Background

A. Related Works

In this section, we summarize the related works and three representative methods for incomplete multi-view semi-supervised classification methods.

In recent years, how to deal with incomplete multi-view data has been studied. Earlier methods are designed to cluster incomplete two-view data. The method [20] aims to learn a latent subspace based on nonnegative matrix factorization, where both view-specific examples and complete examples are homogeneously represented. The methods proposed in [23], [22], [24] are designed based on [20], and employ different regularization terms and constraints. Since these two-view methods usually fail to deal with data which have more than two incomplete views, the methods [21], [25], [32] introduce diagonal weight matrices for multiple views to distinguish their present samples from missing samples. Besides aforementioned matrix factorization-based methods, several kernel-based, graph-based and deep methods [18], [19], [26], [33], [34] have been proposed to cluster incomplete multi-view data. Except for unsupervised methods, some recent studied have been proposed to deal with incomplete multi-view data in supervised [35], [14], [15], [13] and semi-supervised [3], [30] scenarios. SMDDRL [3] designs shared and specific representation learning network to enhance the label information by combining deep metric learning and density clustering. However, SMDDRL takes full data representations as the input and can not process incomplete multi-view data directly. The method [35] focus on partial multi-view classification, which learns common latent representations from the encoding networks, and trains a clustering-like classification model based on representations simultaneously. To classify incomplete multi-view data in semi-supervised scenario, semi-supervised multi-modal learning with incomplete modalities (SLIM) proposed in [30] learns view-specific projection matrices and a common label matrix simultaneously, and reconstruct the incomplete similarity matrices based on the label matrix. However, the reconstruction way may make the label matrix fail to disclose the local structure of the similarity matrices. Besides, SLIM has high computational complexity with both data size and dimensionality, which limits its efficiency.

Under the manifold assumption, graph-based methods achieve promising performance in multi-view semi-supervised classification. The method proposed in [36] fuses the pre-constructed graphs of multiple views by the exponential decay incorporation strategy, and conducts the semi-supervised learning on the fused graph simultaneously. The method proposed in [4] learns sparse weights to fuse linearly combine graphs of multiple views to perform label propagation on the fused graph. The auto-weighted multiple graph learning (AMGL) method proposed in [11] adopts the sqrt root function to integrate different graphs, which assigns weights for different views automatically and has no explicit view weight factors. The adaptive multi-modal semi-supervised classification method proposed in [10] considers each type of features as one modality, and learns a common class indicator matrix, label matrices and weights for each modality simultaneously. The

method proposed in [5] utilizes a parameter-free way to perform multi-view semi-supervised classification, view weight learning and the common graph construction simultaneously. Some recent multi-view semi-supervised feature extraction methods [37], [38] adopt the graph terms as regularization. Although these graph-based methods have achieved good predictive performance, they require each view to be complete, and cannot deal with incomplete multi-view data directly. Besides, the graph construction and the optimization bring high computational burdens w.r.t. data size, which makes them have low efficiency to deal with large-size data.

B. Representative Incomplete Semi-supervised Multi-View Learning Methods

1) Semi-supervised Multi-modal Learning with Incomplete Modalities (SLIM): SLIM is a semi-supervised method for incomplete multi-view data [30]. Based on given label matrix $\mathbf{Y}_l$ and incomplete data matrices $\{\mathbf{X}^{(v)}\}_{v=1}^{V}$, SLIM simultaneously learns a common label matrix $\mathbf{F} \in [0,1]^{n\times C}$ and view-specific transformation matrices $\{\mathbf{W}^{(v)}, \mathbf{b}^{(v)}\}_{v=1}^{V}$ where $\mathbf{W}^{(v)} \in \mathbb{R}^{d^{(v)}\times C}$ and $\mathbf{b}^{(v)} \in \mathbb{R}^{1\times C}$. Besides, SLIM utilizes the corresponding elements of $\mathbf{F}\mathbf{F}^T$ to approximate the incomplete similarity matrices $\{\mathbf{S}^{(v)}\}_{v=1}^{V}$ of multiple views, where $\mathbf{S}^{(v)} \in \mathbb{R}^{n\times n}$. The model can be written as:

$$\begin{cases} \min\limits_{\mathbf{W}^{(v)},\mathbf{F}} \sum\limits_{v=1}^{V} \frac{1}{n^{(v)}} [||\mathbf{O}^{(v)}(\mathbf{X}^{(v)}\mathbf{W}^{(v)} + \mathbf{1}_n\mathbf{b}^{(v)} - \mathbf{F})||_F^2 \\ \quad + \lambda_1||\mathbf{W}^{(v)}||_F^2 + \lambda_2||\mathbf{O}^{(v)}(\mathbf{S}^{(v)} - \mathbf{F}\mathbf{F}^T)\mathbf{O}^{(v)}||_F] \\ s.t.\ \mathbf{F}_l = \mathbf{Y}_l, 0 \leqslant \mathbf{F} \leqslant 1, \end{cases} \tag{1}$$

where $n^{(v)} = \sum_{i=1}^{n} o_{ii}^{(v)}$ is the number of present samples of the v-th view, $\lambda_1 > 0$ and $\lambda_2 > 0$ are two balanced parameters.

2) Adaptive Multi-modal Semi-supervised Classification (AMMSS): AMMSS is a graph-based semi-supervised classification method [10] for complete multi-view data. Based on the independently constructed graph matrices $\{\mathbf{A}^{(v)}\}_{v=1}^{V}$ of multiple views, AMMSS learns a common label matrix $\mathbf{F} \in \mathbb{R}^{n\times C}$, view-specific label matrices $\{\mathbf{F}^{(v)}\}_{v=1}^{V}$ and a weight vector $\boldsymbol{\alpha} = [\alpha_1, ..., \alpha_V]^T \in \mathbb{R}^V$ simultaneously. The optimization problem can be written as:

$$\begin{aligned} &\min_{\mathbf{F},\boldsymbol{\alpha},\mathbf{F}^{(v)}} \sum_{v=1}^{V} \{\alpha_v^{\mu} Tr[(\mathbf{F}^{(v)})^T\mathbf{L}^{(v)}\mathbf{F}^{(v)}] + \lambda||\mathbf{F}^{(v)} - \mathbf{F}||_F^2\} \\ &\quad s.t.\ \mathbf{F}_l = \mathbf{Y}_l, \boldsymbol{\alpha} \in \Delta_1^V \end{aligned} \tag{2}$$

where $\mu \geqslant 1$ and $\lambda > 0$ are two hyper-parameters, $\mathbf{L}^{(v)}$ is the Laplacian matrix of $\mathbf{A}^{(v)}$.

3) Auto-weighted Multiple Graph Learning (AMGL): AMGL is a graph-based semi-supervised classification method [11] for complete multi-view data. First, AMGL constructs graph matrices $\{\mathbf{A}^{(v)}\}_{v=1}^{V}$ on each view individually. Then, AMGL perform label propagation on a implicitly unified graph to learn a common label matrix $\mathbf{F} \in \mathbb{R}^{n\times C}$. The optimization problem can be written as:

$$\min_{\mathbf{F}} \sum_{v=1}^{V} \sqrt{Tr\big(\mathbf{F}^T\mathbf{L}^{(v)}\mathbf{F}\big)}, \quad s.t.\ \mathbf{F}_l = \mathbf{Y}_l \tag{3}$$

The problem (3) integrates the losses of multiple views according to the sqrt root integration strategy.

III. Absent Multi-view Semi-supervised Classification

In this section, we first propose a way to construct partial graph matrices based on anchor strategy, then the optimization problem of the proposed model is introduced.

A. Notations and Problem Setting

We first introduce some notations. Matrices and vectors are written as boldface uppercase letters and boldface lowercase letters, respectively. For a matrix $\mathbf{M} \in \mathbb{R}^{n\times d}$, its ij-th element and i-th row are denoted by m_{ij} and $\mathbf{m}_i$, respectively. $\mathbf{M}^{(v)}$ represents a matrix derived from the v-th view. The transpose, inverse, trace and F-norm of the matrix $\mathbf{M}$ are denoted by $\mathbf{M}^T$, $\mathbf{M}^{-1}$, $Tr(\mathbf{M})$ and $||\mathbf{M}||_F$, respectively. $Ind \overset{\text{def}}{=} \{\mathbf{M} \in \{0,1\}^{n\times C} | \mathbf{M}\mathbf{1}_C = \mathbf{1}_n\}$ is the set of indicator matrix. $\Delta_d^n \overset{\text{def}}{=} \{\mathbf{M} \in \mathbb{R}^{n\times d} | \mathbf{M} \geqslant 0, \mathbf{M}\mathbf{1}_d = \mathbf{1}_n\}$ denotes the probabilistic simplex. For a vector $\mathbf{m}_i$, its 2-norm is denoted by $||\mathbf{m}_i||$.

Assume that a dataset $\mathbf{X} = [\mathbf{x}_1; ...; \mathbf{x}_n]$ has n instances, where $\mathbf{x}_i \in \mathbb{R}^{1\times d}$ is the i-th instance. Each instance has V view representations, i.e., $\mathbf{x}_i = [\mathbf{x}_i^{(1)}, ..., \mathbf{x}_i^{(V)}]$, where $\mathbf{x}_i^{(v)} \in \mathbb{R}^{1\times d^{(v)}}$ and $d = \sum_{v=1}^{V} d^{(v)}$. Denote the v-th view data matrix as $\mathbf{X}^{(v)} = [\mathbf{x}_1^{(v)}; ...; \mathbf{x}_n^{(v)}]$. Supposing that n instances belong to C different classes, the first l instances are labeled while the rest $u = n - l$ instances are unlabeled. Denote $\mathbf{Y} = [\mathbf{y}_1; ...; \mathbf{y}_n] \in \mathbb{R}^{n\times C}$ as the label matrix for all data points, where $\mathbf{y}_i \in \{0,1\}^{1\times C}$ is the label vector for the i-th instance. $\mathbf{Y}_l = [\mathbf{y}_1; \ldots; \mathbf{y}_l] \in \mathbb{R}^{l\times C}$ collects the given label vectors, while $\mathbf{Y}_u = [\mathbf{y}_{l+1}; \ldots; \mathbf{y}_n] \in \mathbb{R}^{u\times C}$ is the label matrix for the u unlabeled instances.

Different from traditional multi-view setting, each $\mathbf{x}_i^{(v)}$ can be absent in incomplete multi-view setting. To distinguish the present and the absent samples, a diagonal indicator matrix $\mathbf{O}^{(v)} \in \{0,1\}^{n\times n}$ for the v-th view is defined as follows:

$$o_{ii}^{(v)} = \begin{cases} 0, & \text{if } \mathbf{x}_i^{(v)} \text{ is absent} \\ 1, & \text{otherwise.} \end{cases} \tag{4}$$

B. Partial Graph Construction

To use the complementary information of multiple views, we construct partial graph matrices on each view. Traditional graph construction methods calculate the similarity between each pair of samples, which makes their efficiency drop rapidly with the increase of data size. To solve this problem, we propose to adopt the following anchor-based strategy.

1) Anchor Generation: To generate $m^{(v)}$ representative anchors $\{\mathbf{u}_j^{(v)}\}_{j=1}^{m^{(v)}}$ on each view individually, where $\mathbf{u}_j^{(v)} \in \mathbb{R}^{1\times d^{(v)}}$ is the j-th anchor of the v-th view, we perform K-means on the present samples of each view by solving the following problem:

$$\min_{\mathbf{G}^{(v)}\in Ind, \mathbf{U}^{(v)}} ||\mathbf{O}^{(v)}(\mathbf{X}^{(v)} - \mathbf{G}^{(v)}\mathbf{U}^{(v)})||_F^2 \tag{5}$$

where $\mathbf{U}^{(v)} = [\mathbf{u}_1^{(v)}; ...; \mathbf{u}_m^{(v)}] \in \mathbb{R}^{m^{(v)}\times d^{(v)}}$ collects the $m^{(v)}$ cluster centers as the v-th view representative anchors, and $\mathbf{G}^{(v)}$ is the cluster indicator matrix. The problem (5) can be solved by updating $\mathbf{U}^{(v)}$ and $\mathbf{G}^{(v)}$ alternatively and iteratively.

2) Representation Matrix Learning: We calculate a representation matrix $\mathbf{Z}^{(v)} \in \mathbb{R}^{n\times m^{(v)}}$ to measure the similarity between samples $\{\mathbf{x}_i^{(v)}\}_{i=1}^n$ and anchors $\{\mathbf{u}_j^{(v)}\}_{j=1}^{m^{(v)}}$ of each view, where $z_{ij}^{(v)}$ reflects the similarity between $\mathbf{x}_i^{(v)}$ and $\mathbf{u}_j^{(v)}$. First, we define the similarity loss as follows:

Defination (Similarity Loss). *Given two vectors* $\mathbf{a}$ *and* $\mathbf{b}$ *with the same dimensionality, and a nonnegative number* c *which measures the similarity between* $\mathbf{a}$ *and* $\mathbf{b}$*, the similarity loss* $\ell(c|\mathbf{a},\mathbf{b})$ *is defined as*

$$\ell(c|\mathbf{a},\mathbf{b}) \overset{def}{=} c||\mathbf{a}-\mathbf{b}||^2. \tag{6}$$

Each $\mathbf{z}_i^{(v)}$ is calculated by solving the following problem:

$$\begin{cases} \min\limits_{\mathbf{z}_i^{(v)}} \sum\limits_{j=1}^{m^{(v)}} \ell(z_{ij}^{(v)}|o_{ii}^{(v)}\mathbf{x}_i^{(v)},\mathbf{u}_j^{(v)}) + \kappa||\mathbf{z}_i^{(v)}||^2 \\ s.t.\ \mathbf{z}_i^{(v)} \geqslant 0, \mathbf{z}_i^{(v)}\mathbf{1}_{m^{(v)}} = o_{ii}^{(v)}, \end{cases} \tag{7}$$

where $\kappa > 0$ is a parameter. When $o_{ii}^{(v)} = 0$, the elements of $\mathbf{z}_i^{(v)}$ are all 0, the similarity information between $\mathbf{x}_i^{(v)}$ and $\{\mathbf{u}_j^{(v)}\}_{j=1}^{m^{(v)}}$ are missing. When $o_{ii}^{(v)} = 1$, the problem (7) has a closed form solution $\mathbf{z}_i^{(v)}$ with n_k nonzero elements [39]. Therefore, only $n^{(v)}$ rows of $\mathbf{Z}^{(v)}$ contain certain information which reflect the similarity between each pair of the v-th view present samples and anchors.

3) Partial Similarity Matrix Construction: After obtaining the representation matrix $\mathbf{Z}^{(v)}$ such that $\mathbf{O}^{(v)}\mathbf{X}^{(v)} \approx \mathbf{Z}^{(v)}\mathbf{U}^{(v)}$, the v-th view partial similarity matrix $\mathbf{S}^{(v)} \in \mathbb{R}^{n\times n}$ can be constructed as follows

$$\mathbf{S}^{(v)} = \mathbf{Z}^{(v)}(\mathbf{E}^{(v)})^{-1}(\mathbf{Z}^{(v)})^T \tag{8}$$

where $\mathbf{E}^{(v)} \in \mathbb{R}^{m^{(v)}\times m^{(v)}}$ is a diagonal matrix in which the j-th entry $e_{jj}^{(v)} = \sum_{i=1}^n z_{ij}^{(v)}$. And for each $\mathbf{S}^{(v)}$, we have the following proposition.

Proposition 1. *Given a representation matrix* $\mathbf{Z}^{(v)}$ *computed by Eq. (7),* $\mathbf{S}^{(v)}$ *computed according to Eq. (8) is symmetric and satisfies* $\mathbf{S}^{(v)}\mathbf{1}_n = \mathbf{Z}^{(v)}\mathbf{1}_{m^{(v)}} = \mathbf{O}^{(v)}\mathbf{1}_n$.

Proof. According to Eq. (8), there is

$$s_{ij}^{(v)} = \mathbf{z}_i^{(v)}(\mathbf{E}^{(v)})^{-1}(\mathbf{z}_j^{(v)})^T = \sum_{l=1}^{m^{(v)}} \frac{z_{il}^{(v)} z_{jl}^{(v)}}{\sum_{k=1}^n z_{kl}^{(v)}} \tag{9}$$

One can be easily verify that $s_{ij}^{(v)} = s_{ji}^{(v)}$, which implies that $\mathbf{S}^{(v)}$ is symmetric. It also can be verified that

$$\sum_{j=1}^n s_{ij}^{(v)} = \sum_{l=1}^{m^{(v)}} \frac{z_{il}^{(v)} \sum_{j=1}^n z_{jl}^{(v)}}{\sum_{k=1}^n z_{kl}^{(v)}} = \sum_{l=1}^{m^{(v)}} z_{il}^{(v)} = o_{ii}^{(v)} \tag{10}$$

which indicates $\sum_{j=1}^n s_{ij}^{(v)} = \sum_{i=1}^n s_{ij}^{(v)} = o_{ii}^{(v)}$. □

Based on Eqs. (7) and (9), since the information of some rows of $\mathbf{Z}^{(v)}$ are missing, $s_{ij}^{(v)}$ can reflect the similarity between $\mathbf{x}_i^{(v)}$ and $\mathbf{x}_j^{(v)}$ only if the information of both $\mathbf{z}_i^{(v)}$ and $\mathbf{z}_j^{(v)}$ are complete, in other words, both $o_{ii}^{(v)}$ and $o_{jj}^{(v)}$ are equal to 1. Therefore, $\mathbf{S}^{(v)}$ has certain $n^{(v)} \times n^{(v)}$ elements, which measure the similarity between each pair of the v-th view present samples.

C. Model of AMSC

Denote the v-th view label matrix as $\mathbf{F}^{(v)} = [\mathbf{f}_1^{(v)};...;\mathbf{f}_n^{(v)}] \in \mathbb{R}^{n\times C}$, where $\mathbf{f}_i^{(v)} \in \mathbb{R}^{1\times C}$ is the label vector of $\mathbf{x}_i^{(v)}$. Since the label vectors of $\{\mathbf{x}_i^{(v)}\}_{i=1}^l$ are given, to learn the label vectors of the rest samples, the label propagation based on the v-th view partial similarity matrix $\mathbf{S}^{(v)}$ can be formulated as

$$\min_{\mathbf{F}^{(v)}} \sum_{i=1}^n \sum_{j=1}^n \ell(o_{ii}^{(v)} s_{ij}^{(v)} o_{jj}^{(v)}|\mathbf{f}_i^{(v)},\mathbf{f}_j^{(v)}) \quad s.t.\ \mathbf{F}_l^{(v)} = \mathbf{Y}_l \tag{11}$$

where $\mathbf{F}_l^{(v)}$ denotes the first l rows of $\mathbf{F}^{(v)}$. According to label propagation theory, the label information can propagate from labeled data to unlabeled data only if they are on a same connected component of the graph matrix. Therefore, $o_{ii}^{(v)} s_{ij}^{(v)} o_{jj}^{(v)}$ ensures that the label information propagates on the $n^{(v)} \times n^{(v)}$ certain elements of $\mathbf{S}^{(v)}$. When constructing the partial similarity matrix $\mathbf{S}^{(v)}$ by (8), there is

$$o_{ii}^{(v)} s_{ij}^{(v)} o_{jj}^{(v)} = s_{ij}^{(v)} \tag{12}$$

Since every view may suffer from some unlabeled samples missing, the classification results of all unlabeled data can not be obtained by applying (11) on one view. Besides, if (11) is performed on each view independently, the obtained $\{\mathbf{F}^{(v)}\}_{v=1}^V$ usually have ambiguous classification results.

To get unambiguous classification results of all unlabeled data points based on information of multiple views, we learn a common probability label matrix $\mathbf{F} \in \Delta_C^n$ together with view-specific probability label matrices $\{\mathbf{F}^{(v)}\}_{v=1}^V$. To establish a reasonable learning mechanism, for each view-specific label vector $\mathbf{f}_i^{(v)}$, based on Eq. (11), we define its intra-view similarity loss as follows:

• *Intra-view similarity loss:*

$$\sum_{j=1}^n \ell(o_{ii}^{(v)} s_{ij}^{(v)} o_{jj}^{(v)}|\mathbf{f}_i^{(v)},\mathbf{f}_j^{(v)}) = \sum_{j=1}^n s_{ij}^{(v)}||\mathbf{f}_i^{(v)} - \mathbf{f}_j^{(v)}||^2 \tag{13}$$

The intra-view similarity loss of $\mathbf{f}_i^{(v)}$ measures the similarity losses between $\mathbf{f}_i^{(v)}$ and $\{\mathbf{f}_i^{(v)}\}_{i=1}^n$ based on similarity information contained in $\mathbf{s}_i^{(v)}$. To make $\{\mathbf{F}^{(v)}\}_{v=1}^V$ interact with $\mathbf{F}$ reasonably, we introduce class indicator vectors $\{\mathbf{t}_{(c)}\}_{c=1}^C$, and define the extra-view similarity loss of $\mathbf{f}_i^{(v)}$ as follows:

• *Extra-view similarity loss:*

$$\sum_{c=1}^C \ell(o_{ii}^{(v)} f_{ic}^\gamma|\mathbf{f}_i^{(v)},\mathbf{t}_{(c)}) = \sum_{c=1}^C o_{ii}^{(v)} f_{ic}^\gamma||\mathbf{f}_i^{(v)} - \mathbf{t}_{(c)}||^2 \tag{14}$$

where $\mathbf{t}_{(c)} \in \{0,1\}^{1\times C}$ is the coding vector of the c-th class, $\gamma \geqslant 1$ is an adaptive parameter, f_{ic} is the (i,c)-th element of $\mathbf{F}$, and $f_{ic}^\gamma = (f_{ic})^\gamma$. For $\mathbf{t}_{(c)}$, only its c-th element is 1 and the other ones are 0. The extra-view similarity loss of $\mathbf{f}_i^{(v)}$ sums the similarity losses between $\mathbf{f}_i^{(v)}$ and $\{\mathbf{t}_c\}_{c=1}^C$ based on $\{o_{ii}^{(v)} f_{ic}^\gamma\}_{c=1}^C$. Since $\{\mathbf{t}_c\}_{c=1}^C$ are class indicator vectors, minimizing the extra-view similarity loss of $\mathbf{f}_i^{(v)}$ enables $\mathbf{f}_i^{(v)}$ to have clear classification structure. Besides, the defined extra-view similarity loss has an instance weight learning mechanism. For $\mathbf{f}_i^{(v)}$, its contribution importance of

intra-view similarity loss and extra-view similarity loss are denoted by $\rho_i^{(v)}$ and $\varrho_i^{(v)}$, respectively, and they are defined as

$$\rho_i^{(v)} = \sum_{j=1}^{n} s_{ij}^{(v)}, \quad \varrho_i^{(v)} = o_{ii}^{(v)} \sum_{c=1}^{C} f_{ic}^{\gamma}. \tag{15}$$

According to Eq. (10), there is $\rho_i^{(v)} = o_{ii}^{(v)}$, which means $\{\rho_i^{(v)}\}_{i=1}^n$ distinguishes the contribution importance of missing samples and present samples. When $\gamma = 1$, $\varrho_i^{(v)} = o_{ii}^{(v)}$ has the similar effect with $\rho_i^{(v)}$. When $1 < \gamma < \infty$, $\{\varrho_i^{(v)}\}_{i=1}^n$ can further distinguish different types of present samples according to the classification certainty $\sum_{c=1}^{C} f_{ic}^{\gamma}$, which ensures the significance of clearly classified present samples on each view, and suppresses the importance of uncertain samples on the training procedure automatically.

Since $n^{(v)}$ samples are present on the v-th view, the mean loss $J(\mathbf{F}^{(v)}, \mathbf{F})$ of the v-th view can be calculated by

$$\begin{aligned} &J(\mathbf{F}^{(v)}, \mathbf{F}) = \\ &\sum_{i=1}^{n} \frac{o_{ii}^{(v)}}{n^{(v)}} [\sum_{j=1}^{n} \ell(s_{ij}^{(v)} | \mathbf{f}_i^{(v)}, \mathbf{f}_j^{(v)}) + \sum_{c=1}^{C} \ell(f_{ic}^{\gamma} | \mathbf{f}_i^{(v)}, \mathbf{t}_{(c)})]. \end{aligned} \tag{16}$$

After proposing the objective functions of each view, a rough way to obtain the multi-view formulation is to add up them directly. However, this way neglects the different importance of multiple views. To consider their different contributions, in this paper, we incorporate multiple views by the p-th root integration strategy [31]. Besides, in order to enable $\{\mathbf{F}^{(v)}\}_{v=1}^V$ to capture more diversity information of multiple views, we constrain $\mathbf{F}_l = \mathbf{Y}_l$ rather than $\mathbf{F}_l^{(v)} = \mathbf{Y}_l$. As a result, we propose AMSC as follows:

$$\begin{cases} \min\limits_{\mathbf{F}, \{\mathbf{F}^{(v)}\}_{v=1}^V} \sum\limits_{v=1}^{V} J(\mathbf{F}^{(v)}, \mathbf{F})^p = \\ \sum\limits_{v=1}^{V} \sqrt[\frac{1}{p}]{\sum\limits_{i=1}^{n} \frac{o_{ii}^{(v)}}{n^{(v)}} [\sum\limits_{j=1}^{n} \ell(s_{ij}^{(v)} | \mathbf{f}_i^{(v)}, \mathbf{f}_j^{(v)}) + \sum\limits_{c=1}^{C} \ell(f_{ic}^{\gamma} | \mathbf{f}_i^{(v)}, \mathbf{t}_{(c)})]} \\ s.t.\ \mathbf{F}_l = \mathbf{Y}_l, \mathbf{F} \in \Delta_C^n, \mathbf{F}^{(v)} \in \Delta_C^n, (\forall v), \end{cases} \tag{17}$$

where $0 < p \leq 1$ is a hyper-parameter. The p-th root strategy distinguishes the importance of various views according to their losses, which enables the views with smaller losses to play more important roles in learning stage.

IV. Optimization

In this section, we first introduce the optimization procedures of AMSC algorithm. Then we discuss the convergence behavior of the proposed AMSC algorithm. Lastly, we analyze its computational complexity.

A. Algorithm

The p-th root integration strategy makes the problem (17) difficult to be solved directly. Previous works [11], [40], [41], [42] which adopt sqrt integration strategy use re-weighted method [43] for optimization. Different from them, we obtain solution of (17) by solving the following problem with a explicit weight vector $\boldsymbol{\alpha} = [\alpha_1, ..., \alpha_V]^T \in \mathbb{R}^V$:

$$\begin{aligned} &\min_{\mathbf{F}, \{\mathbf{F}^{(v)}\}_{v=1}^V, \boldsymbol{\alpha}} \sum_{v=1}^{V} \alpha_v^{\mu} J(\mathbf{F}^{(v)}, \mathbf{F}) = \\ &\sum_{v=1}^{V} \frac{\alpha_v^{\mu}}{n^{(v)}} \sum_{i=1}^{n} [\sum_{j=1}^{n} \ell(s_{ij}^{(v)} | \mathbf{f}_i^{(v)}, \mathbf{f}_j^{(v)}) + \sum_{c=1}^{C} \ell(f_{ic}^{\gamma} o_{ii}^{(v)} | \mathbf{f}_i^{(v)}, \mathbf{t}_{(c)})] \\ &s.t.\ \mathbf{F}_l = \mathbf{Y}_l, \mathbf{F} \in \Delta_C^n, \boldsymbol{\alpha} \in \Delta_V^1, \mathbf{F}^{(v)} \in \Delta_C^n, (\forall v), \end{aligned} \tag{18}$$

where $\mu = \frac{p-1}{p} \leqslant 0$ is a exponential decay hyper-parameter. Different from previous works [36], [44], [45] with $\mu \geqslant 1$, (18) requires $\mu \leqslant 0$. To solve (18), an alternative strategy is adopted to optimize three different groups of variables iteratively.

1) Updating View-specific Label Matrices $\{\mathbf{F}^{(v)}\}_{v=1}^V$: With $\mathbf{F}$ and $\boldsymbol{\alpha}$ fixed, the problem (18) disassembles into V individual and separate subproblems. By removing the constant terms, the v-th subproblem about $\mathbf{F}^{(v)}$ can be written as the following matrix form:

$$\begin{aligned} \min_{\mathbf{F}^{(v)} \in \Delta_C^n} \ & 2Tr[(\mathbf{O}^{(v)}\mathbf{F}^{(v)})^T(\mathbf{O}^{(v)} - \mathbf{S}^{(v)})\mathbf{O}^{(v)}\mathbf{F}^{(v)}] \\ & + Tr[(\mathbf{O}^{(v)}\mathbf{F}^{(v)})^T(\mathbf{D}\mathbf{O}^{(v)}\mathbf{F}^{(v)} - 2\mathbf{M})], \end{aligned} \tag{19}$$

where $\mathbf{D} \in \mathbb{R}^{n \times n}$ is a diagonal matrix with i-th entry $d_{ii} = \sum_{c=1}^{C} f_{ic}^{\gamma}$; $\mathbf{M} = \mathbf{F}^{\gamma}$ is computed in element wise. By analyzing (19), it can be observed that the i-th row $\mathbf{f}_i^{(v)}$ of $\mathbf{F}^{(v)}$ needs to be optimized only if $o_{ii}^{(v)} = 1$. For $\mathbf{F}^{(v)}$, $\mathbf{M}$ and $\mathbf{Z}^{(v)}$, their $n^{(v)}$ rows corresponding to the v-th view present samples are collected by $\mathbf{F}_{\Omega}^{(v)}$, $\mathbf{M}_{\Omega}^{(v)}$ and $\mathbf{Z}_{\Omega}^{(v)}$, respectively. The $n^{(v)} \times n^{(v)}$ elements of $\mathbf{S}^{(v)}$ which measure the similarity between each pair of present samples are collected by $\mathbf{S}_{\Omega}^{(v)}$. The corresponding $n^{(v)} \times n^{(v)}$ elements of $\mathbf{D}$ are collected by $\mathbf{D}_{\Omega}^{(v)}$. As a result, the problem (19) can be transformed as:

$$\min_{\mathbf{F}_{\Omega}^{(v)} \in \Delta_C^{n^{(v)}}} Tr\{(\mathbf{F}_{\Omega}^{(v)})^T[(\mathbf{P}^{(v)} - 2\mathbf{S}_{\Omega}^{(v)})\mathbf{F}_{\Omega}^{(v)} - 2\mathbf{M}_{\Omega}^{(v)}]\}, \tag{20}$$

where $\mathbf{P}^{(v)} = 2\mathbf{I}_{n^{(v)}} + \mathbf{D}_{\Omega}^{(v)}$ is a diagonal matrix. To solve the problem (17), we have the following proposition.

Proposition 2. *The problem (20) shares the same optimal solution with the following problem*

$$\min_{\mathbf{F}_{\Omega}^{(v)}} Tr\{(\mathbf{F}_{\Omega}^{(v)})^T[(\mathbf{P}^{(v)} - 2\mathbf{S}_{\Omega}^{(v)})\mathbf{F}_{\Omega}^{(v)} - 2\mathbf{M}_{\Omega}^{(v)}]\}. \tag{21}$$

By setting the derivative of the objective function in Eq. (21) w.r.t. $\mathbf{F}_{\Omega}^{(v)}$ as zero, the optimal $\mathbf{F}_{\Omega}^{(v)}$ of (20) and (21) is

$$\mathbf{F}_{\Omega}^{(v)} = (\mathbf{P}^{(v)} - 2\mathbf{S}_{\Omega}^{(v)})^{-1}\mathbf{M}_{\Omega}^{(v)}. \tag{22}$$

Proof. Since the problem (21) is convex, $\mathbf{F}_{\Omega}^{(v)}$ obtained by (22) is the optimal solution of (21). Since (20) and (21) have the same objective function, if $\mathbf{F}_{\Omega}^{(v)}$ satisfies $\mathbf{F}_{\Omega}^{(v)} \in \Delta_1^V$ in (22), it is also the optimal solution of (20).

According to the definition of $\mathbf{S}_{\Omega}^{(v)}$, there is $\mathbf{S}_{\Omega}^{(v)}\mathbf{1}_{n^{(v)}} = (\mathbf{S}_{\Omega}^{(v)})^T\mathbf{1}_{n^{(v)}} = \mathbf{I}_{n^{(v)}}\mathbf{1}_{n^{(v)}}$. Since $\mathbf{D}$ is a diagonal matrix with $d_{ii} > 0$ and $\mathbf{P}^{(v)} = 2\mathbf{I}_{n^{(v)}} + \mathbf{D}_{\Omega}^{(v)}$, $(\mathbf{P}^{(v)} - 2\mathbf{S}_{\Omega}^{(v)})$ is a symmetric and strictly diagonally dominant matrix. By analyzing

the algebraic cofactor of each element of $(\mathbf{P}^{(v)} - 2\mathbf{S}_{\Omega}^{(v)})$, it is to check that $(\mathbf{P}^{(v)} - 2\mathbf{S}_{\Omega}^{(v)})^{-1} \geqslant 0$. On the other hand, since $\mathbf{M} = \mathbf{F}^{\gamma}$ is computed in element wise, $\mathbf{M}_{\Omega}^{(v)}$ satisfies $\mathbf{M}_{\Omega}^{(v)} \geqslant 0$. Therefore, $\mathbf{F}_{\Omega}^{(v)} = (\mathbf{P}^{(v)} - 2\mathbf{S}_{\Omega}^{(v)})^{-1}\mathbf{M}_{\Omega}^{(v)} \geqslant 0$.

According to the definitions of $\mathbf{D}$ and $\mathbf{M}$, we have

$$\mathbf{M}_{\Omega}^{(v)}\mathbf{1}_C = \mathbf{D}_{\Omega}^{(v)}\mathbf{1}_{n^{(v)}}. \tag{23}$$

Since $\mathbf{I}_{n^{(v)}}\mathbf{1}_{n^{(v)}} = \mathbf{S}_{\Omega}^{(v)}\mathbf{1}_{n^{(v)}}$, there is

$$\mathbf{0}_{n^{(v)}} = 2(\mathbf{I}_{n^{(v)}} - \mathbf{S}_{\Omega}^{(v)})\mathbf{1}_{n^{(v)}}. \tag{24}$$

Summing Eq. (23) and Eq. (24) in the two sides, we arrive at

$$\begin{aligned} &\mathbf{M}_{\Omega}^{(v)}\mathbf{1}_C = (2\mathbf{I}_{n^{(v)}} + \mathbf{D}_{\Omega}^{(v)} - 2\mathbf{S}_{\Omega}^{(v)})\mathbf{1}_{n^{(v)}} \\ \Rightarrow &(\mathbf{P}^{(v)} - 2\mathbf{S}_{\Omega}^{(v)})^{-1}\mathbf{M}_{\Omega}^{(v)}\mathbf{1}_C = \mathbf{1}_{n^{(v)}}. \end{aligned} \tag{25}$$

Therefore, $\mathbf{F}_{\Omega}^{(v)}\mathbf{1}_C = \mathbf{1}_{n^{(v)}}$. □

If we simply use Eq. (22) derived from Eq. (21) to update $\mathbf{F}_{\Omega}^{(v)}$, the computational complexity is cubic with respect to the present data size $n^{(v)}$. To make the proposed algorithm more efficient, we propose another way to compute $\mathbf{F}_{\Omega}^{(v)}$, which scales linearly in time with respect to $n^{(v)}$. Specifically, according to Eq. (7) and Eq. (8), $\mathbf{S}_{\Omega}^{(v)} = \mathbf{Z}_{\Omega}^{(v)}(\mathbf{E}^{(v)})^{-1}(\mathbf{Z}_{\Omega}^{(v)})^T$. The woodbury identity matrix [46] can be written as

$$\begin{aligned} &(\mathbf{A} + \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^T)^{-1} \\ &= \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{B}(\mathbf{C} + \mathbf{B}^T\mathbf{A}^{-1}\mathbf{B})^{-1}\mathbf{B}^T\mathbf{A}^{-1}. \end{aligned} \tag{26}$$

According to (26), $\mathbf{F}_{\Omega}^{(v)}$ can also be calculated as

$$\begin{aligned} \mathbf{F}_{\Omega}^{(v)} =&(\mathbf{P}^{(v)})^{-1}\mathbf{M}_{\Omega}^{(v)} + (\mathbf{P}^{(v)})^{-1}\mathbf{Z}_{\Omega}^{(v)}[\frac{\mathbf{E}^{(v)}}{2} \\ &-(\mathbf{Z}_{\Omega}^{(v)})^T(\mathbf{P}^{(v)})^{-1}\mathbf{Z}_{\Omega}^{(v)}]^{-1}(\mathbf{Z}_{\Omega}^{(v)})^T(\mathbf{P}^{(v)})^{-1}\mathbf{M}_{\Omega}^{(v)}. \end{aligned} \tag{27}$$

Therefore, the inverse of the $n^{(v)} \times n^{(v)}$ matrix in Eq. (22) can be substituted by the inverse of a diagonal matrix, the inverse of a $m^{(v)} \times m^{(v)}$ matrix and some multiplications.

2) Updating Consensus Label Matrix $\mathbf{F}$*:* With $\boldsymbol{\alpha}$ and $\{\mathbf{F}^{(v)}\}_{v=1}^{V}$ fixed, the relations among instances are decoupled. Since $\mathbf{F}_l = \mathbf{Y}_l$, only $\mathbf{F}_u$ needs to be updated by solving the following $u(i = l+1, ..., n)$ problems individually:

$$\min_{\mathbf{f}_i \in \Delta_C^1} \sum_{c=1}^{C} f_{ic}^{\gamma} \sum_{v=1}^{V} \frac{\alpha_v^{\mu} o_{ii}^{(v)}}{n^{(v)}} ||\mathbf{f}_i^{(v)} - \mathbf{t}_{(c)}||^2. \tag{28}$$

Denote $q_{ic} = \sum_{v=1}^{V} \alpha_v^{\mu} o_{ii}^{(v)} ||\mathbf{f}_i^{(v)} - \mathbf{t}_{(c)}||^2 / n^{(v)}$. When $\gamma = 1$, (28) is a sparse problem, and the optimal solution is

$$f_{ij} = \left\langle j = arg \min_{c \in [1,C]} q_{ic} \right\rangle, \tag{29}$$

where $< \cdot >$ is 1 if the argument is true or 0 otherwise. When $\gamma > 1$, by analyzing the Lagrange function of (28) and combing the constraint, the optimal solution of (28) is

$$f_{ic} = \frac{\left(q_{ic}\right)^{\frac{1}{1-\gamma}}}{\sum_{c=1}^{C} \left(q_{ic}\right)^{\frac{1}{1-\gamma}}}. \tag{30}$$

3) Updating Vector $\boldsymbol{\alpha}$*:* With $\{\mathbf{F}^{(v)}\}_{v=1}^{V}$ and $\mathbf{F}$ fixed, $\{J(\mathbf{F}^{(v)}, \mathbf{F})\}_{v=1}^{V}$ can be computed accordingly, then we can update $\boldsymbol{\alpha}$ by solving the following problem:

$$\min_{\boldsymbol{\alpha} \in \Delta_V^1} \sum_{v=1}^{V} \alpha_v^{\mu} J(\mathbf{F}^{(v)}, \mathbf{F}). \tag{31}$$

When $\mu = 0$, i.e., $p = 1$, we can fix $\alpha_v = 1/V$. When $\mu < 0$, the optimal solution of (31) is

$$\alpha_v = \frac{\left[J(\mathbf{F}^{(v)}, \mathbf{F})\right]^{\frac{1}{1-\mu}}}{\sum_{v=1}^{V} \left[J(\mathbf{F}^{(v)}, \mathbf{F})\right]^{\frac{1}{1-\mu}}}. \tag{32}$$

By the above three steps, we alternatively update $\{\mathbf{F}^{(v)}\}_{v=1}^{V}$, $\mathbf{F}$ as well as $\boldsymbol{\alpha}$, and repeat these procedures iteratively until the objective of (17) converges. Since the alternative optimization strategy is adopted, we initialize $f_{ic} = 1/C$ $(i = l+1, ..., n; c = 1, ..., C)$ and $\alpha_v = 1/V$ $(v = 1, ..., V)$. At last, we resort to a decision function to assign the single class label $\mathbf{y}_i$ for each unlabeled instance:

$$y_{ij} = \left\langle j = arg \max_{c \in [1,C]} f_{ic} \right\rangle. \tag{33}$$

The complete algorithm can be summarized in Alg. 1.

Algorithm 1 Optimization of AMSC.

Input: Incomplete V view data $\{\mathbf{X}^{(v)}\}_{v=1}^{V}$, indicator matrices $\{\mathbf{O}^{(v)}\}_{v=1}^{V}$, label matrix $\mathbf{Y}_l$, $m^{(v)}$, p and γ.
Initialization:
1. Obtain $\{\mathbf{U}^{(v)}\}_{v=1}^{V}$ by solving (5).
2. Compute $\{\mathbf{Z}^{(v)}\}_{v=1}^{V}$ by solving (7).
3. Initialize each element of $\mathbf{F}_u$ and $\boldsymbol{\alpha}$ as $1/C$ and $1/V$.
While *not converged* **do**
1: Calculate $\mathbf{D}$ and $\mathbf{M}$, and update each $\mathbf{F}^{(v)}$ by Eq. (27).
2: Compute $\mathbf{Q}$, and update $\mathbf{F}_u$ by Eq. (29) or (30).
3: Calculate each $J(\mathbf{F}^{(v)}, \mathbf{F})$, and update $\boldsymbol{\alpha}$ by Eq. (32).
End While
Output: The discrete label matrix $\mathbf{Y}_u$ with Eq. (33).

B. Convergence Analysis

Proposition 3. *Alg. 1 will monotonically decrease the objective of the problem (18), which makes the solution converge to a stationary point of the problem (18).*

Proof. Suppose that the updated $\mathbf{F}^{(v)}$, $\mathbf{F}$ and $\boldsymbol{\alpha}$ are $\tilde{\mathbf{F}}^{(v)}$, $\tilde{\mathbf{F}}$ and $\tilde{\boldsymbol{\alpha}}$, respectively. In Alg.1, the original problem (18) is divided into three groups of subproblems and each subproblem is a convex problem. Since $\tilde{\mathbf{F}}^{(v)}$, $\tilde{\mathbf{F}}$ and $\tilde{\boldsymbol{\alpha}}$ are the optimal solutions of the corresponding subproblems, Alg. 1 decreases the objective of (18) alternatively and iteratively. Since the objective of (18) is lower bounded by 0, Alg. 1 will converge. Denote $\{\hat{\mathbf{F}}^{(v)}\}_{v=1}^{V}$, $\hat{\mathbf{F}}$ and $\hat{\boldsymbol{\alpha}}$ as the converged $\{\mathbf{F}^{(v)}\}_{v=1}^{V}$, $\mathbf{F}$ and $\boldsymbol{\alpha}$ respectively. In the convergence, $\{\{\hat{\mathbf{F}}^{(v)}\}_{v=1}^{V}, \hat{\mathbf{F}}, \hat{\boldsymbol{\alpha}}\}$ satisfies the KKT conditions of the problem (18), therefore, it is at least a stationary point of the problem (18). □

Proposition 4. *Each* $\{\{\tilde{\mathbf{F}}^{(v)}\}_{v=1}^{V}, \tilde{\mathbf{F}}\}$ *in Alg. 1 will monotonically decrease the objective of problem (17) in each iteration, and* $\{\{\hat{\mathbf{F}}^{(v)}\}_{v=1}^{V}, \hat{\mathbf{F}}\}$ *is at least a stationary point of (17).*

Proof. According to Proposition 3, we can conclude that $\sum_{v=1}^{V}\alpha_v^{\mu}J(\tilde{\mathbf{F}}^{(v)},\tilde{\mathbf{F}})\leq\sum_{v=1}^{V}\alpha_v^{\mu}J(\mathbf{F}^{(v)},\mathbf{F})$ in each iteration. Combining $\alpha_v=\left[J(\mathbf{F}^{(v)},\mathbf{F})\right]^{\frac{1}{1-\mu}}/\sum_{v=1}^{V}\left[J(\mathbf{F}^{(v)},\mathbf{F})\right]^{\frac{1}{1-\mu}}$ and $\mu=\frac{p-1}{p}$, it can be inferred that

$$\sum_{v=1}^{V}J(\mathbf{F}^{(v)},\mathbf{F})^{p-1}J(\tilde{\mathbf{F}}^{(v)},\tilde{\mathbf{F}})\leq\sum_{v=1}^{V}J(\mathbf{F}^{(v)},\mathbf{F})^{p}. \tag{34}$$

On the other hand, we define a function $f(x)=x^p$, and then we have $f(J(\mathbf{F}^{(v)},\mathbf{F}))=J(\mathbf{F}^{(v)},\mathbf{F})^p$. When $0<p<1$, $f(J(\mathbf{F}^{(v)},\mathbf{F}))$ is a concave function in the domain of $J(\mathbf{F}^{(v)},\mathbf{F})$. Since the supergradient of $f(x)$ is $f'(x)=px^{p-1}$, then we have $f'(J(\mathbf{F}^{(v)},\mathbf{F}))=pJ(\mathbf{F}^{(v)},\mathbf{F})^{p-1}$. According to the definition of supergradient of a concave function, the following inequality holds:

$$\frac{f(J(\tilde{\mathbf{F}}^{(v)},\tilde{\mathbf{F}}))-f(J(\mathbf{F}^{(v)},\mathbf{F}))}{J(\tilde{\mathbf{F}}^{(v)},\tilde{\mathbf{F}})-J(\mathbf{F}^{(v)},\mathbf{F})}\leq f'(J(\mathbf{F}^{(v)},\mathbf{F})). \tag{35}$$

Thus, we have

$$\begin{aligned}&\sum_{v=1}^{V}J(\tilde{\mathbf{F}}^{(v)},\tilde{\mathbf{F}})^{p}-\sum_{v=1}^{V}pJ(\mathbf{F}^{(v)},\mathbf{F})^{p-1}J(\tilde{\mathbf{F}}^{(v)},\tilde{\mathbf{F}})\\&\quad\leq\sum_{v=1}^{V}J(\mathbf{F}^{(v)},\mathbf{F})^{p}-\sum_{v=1}^{V}pJ(\mathbf{F}^{(v)},\mathbf{F})^{p}\end{aligned} \tag{36}$$

Based on (34) and (36), we arrive at

$$\sum_{v=1}^{V}J(\tilde{\mathbf{F}}^{(v)},\tilde{\mathbf{F}})^{p}\leq\sum_{v=1}^{V}J(\mathbf{F}^{(v)},\mathbf{F})^{p}. \tag{37}$$

Thus, $\{\{\tilde{\mathbf{F}}^{(v)}\}_{v=1}^{V},\tilde{\mathbf{F}}\}$ will monotonically decrease the objective of the problem (17) in each iteration.

Since $\{\{\hat{\mathbf{F}}^{(v)}\}_{v=1}^{V},\hat{\mathbf{F}},\hat{\boldsymbol{\alpha}}\}$ is a stationary point of the problem (18), by replacing $\{\alpha_v\}_{v=1}^{V}$ according to Eq. (32), $\{\{\hat{\mathbf{F}}^{(v)}\}_{v=1}^{V},\hat{\mathbf{F}}\}$ is a stationary point of the following problem:

$$\begin{cases}\min\limits_{\{\mathbf{F}^{(v)}\}_{v=1}^{V},\mathbf{F}}\left[\sum\limits_{v=1}^{V}J(\mathbf{F}^{(v)},\mathbf{F})^{p}\right]^{\frac{1}{p}}\\ s.t.\ \mathbf{F}_l=\mathbf{Y}_l,\mathbf{F}\in\Delta_C^n,\mathbf{F}^{(v)}\in\Delta_C^n\ (\forall v)\end{cases} \tag{38}$$

When $0<p\leqslant 1$, the problem (38) and the problem (17) satisfy the same KKT conditions. Thus, $\{\{\hat{\mathbf{F}}^{(v)}\}_{v=1}^{V},\hat{\mathbf{F}}\}$ is also a stationary point of the problem (17). □

C. Computational Complexity

The partial graph construction has 2 sequential steps: (1) We need $O(\sum_{v=1}^{V}n^{(v)}d^{(v)}m^{(v)}t^{(v)})$ to obtain $\{\mathbf{U}^{(v)}\}_{v=1}^{V}$, where $t^{(v)}$ is the number of iterations of K-means on the v-th view; (2) We need $O(\sum_{v=1}^{V}[n^{(v)}d^{(v)}m^{(v)}+n^{(v)}m^{(v)}\log(m^{(v)})])$ to compute $\{\mathbf{Z}^{(v)}\}_{v=1}^{V}$ and $\{\mathbf{E}^{(v)}\}_{v=1}^{V}$.

The optimization of the problem (18) can be concluded as three alternative and iterative steps: (1) We need $O(\sum_{v=1}^{V}[n^{(v)}m^{(v)}(C+m^{(v)})+(m^{(v)})^3])$ to update $\{\mathbf{F}^{(v)}\}_{v=1}^{V}$ according to Eq. (27); (2) We need $O(\sum_{v=1}^{V}n^{(v)}C+uC)$ to obtain $\mathbf{F}$ by (29) or (30); (3) We need $O(\sum_{v=1}^{V}n^{(v)}m^{(v)}C)$ to update $\boldsymbol{\alpha}$ by Eq. (32).

Since $C\ll m^{(v)}$ and $m^{(v)}\leqslant n^{(v)}$, the overall computational complexity of Alg. 1 is $O(\sum_{v=1}^{V}[n^{(v)}m^{(v)}d^{(v)}t^{(v)}+n^{(v)}(m^{(v)})^2T])$, where T is the number of iteration of Alg. 1.

If adopting one of traditional ways such as [39] to construct partial graph matrices, the computational complexity becomes $O(\sum_{v=1}^{V}(n^{(v)})^2d^{(v)})$. Besides, in each iteration, $\{\mathbf{F}^{(v)}\}_{v=1}^{V}$ need to be updated according to Eq. (22), and the computational complexity becomes $O(\sum_{v=1}^{V}[(n^{(v)})^3+(n^{(v)})^2C])$. Therefore, the overall computational complexity becomes $O(\sum_{v=1}^{V}[(n^{(v)})^2d^{(v)}+(n^{(v)})^3T])$. When $m^{(v)}\ll n^{(v)}$, this optimization procedure has significantly lower efficiency than the proposed Alg. 1.

V. Experiments

In this section, we mainly conduct the extensive experiments to evaluate the performance of our proposed AMSC.

A. Dataset Description

We conduct experiments on 9 datasets, i.e., MSRC-v1[1], Caltech101[2], Protein[3], NBA-NASCAR Sport[4] (NNSpt), Digits[5], Indoor[6], Scene15[7], Kinect Skeleton Action dataset[8] (KSA), MNIST8M[9]. The detailed descriptions of these datasets are listed as follows.

(1) MSRC-v1 consists 240 images belonging to 8 classes. Following [47], we select 7 classes, including *tree*, *building*, *airplane*, *cow*, *face*, *car*, *bicycle*, and each class has 30 images. Six features are extracted, *i.e.*, 256 Local Binary Pattern (LBP), 100 Histogram of Oriented Gradient (HOG), 512 GIST, 1302 CENTRIST, 48 Color Moment (CMT) and 200 SIFT.

(2) Caltech101 is composed of 8677 objective images belonging to 101 categories. Following [48], 7 widely used classes are selected, including *Dolla-Bill*, *Faces*, *Garfield*, *Motorbikes*, *Snoopy*, *Stop-Sign* and *Windsor-Chair*. The subset is named as Caltech7 with 441 images. For each image, the same features with MSRC-v1 are extracted.

(3) Protein consists 629 yeast proteins and is divided into 2 classes: 497 membrane proteins and 132 ribosomal proteins [49]. Each protein is represented by 4910 FFT features, 441 GE features, and 3735 Pfam features.

(4) NNSpt contains 840 images and is divided into 2 classes: 420 NBA images and 420 NASCAR images [50]. Each image has an attached short text. The image is normalized to have 1024 gray features, and 296 TFIDF features have been extracted from each text.

(5) Digits contains 2,000 data points for 0 to 9 ten digit classes, and each class has 200 data points [51]. There are Six public features available: 76 Fourier coefficients of the character shapes (FOU), 216 profile correlations (FAC), 64 Karhunen-love coefficients (KAR), 240 pixel averages in 2×3

[1]https://www.microsoft.com/en-us/research/project/imageunderstanding/
[2]http://www.vision.caltech.edu/Image Datasets/Caltech101/
[3]https://noble.gs.washington.edu/proj/sdp-svm/
[4]http://www.cst.ecnu.edu.cn/ slsun/software/MvLapSVMcode.zip
[5]https://archive.ics.uci.edu/ml/datasets/Multiple+Features
[6]http://web.mit.edu/torralba/www/indoor.html
[7]https://figshare.com/articles/15-Scene_Image_Dataset/7007177
[8]http://www.cs.cmu.edu/ kevinma/
[9]https://www.csie.ntu.edu.tw/ cjlin/libsvmtools/datasets/multiclass.html

TABLE I: Acc and F-score comparisons on 9 datasets with PER= 0. Standard deviation (%) is in the parentheses. The first highest score is in bold. Symbols '•/⊙/∘' denote that AMSC is better/tied/worse than the corresponding method by the paired t-test with confidence level 0.05, respectively.

Dataset	MSRC-v1	Caltech7	Protein	NNSpt	Digits	Indoor	Scene15	KSA	MNIST
Metric	Acc								
LP$_{vote}$	82.4(2.0)•	85.1(1.9)•	88.8(2.9)•	83.9(5.2)•	95.7(0.4)•	67.7(2.9)•	61.4(0.9)•	94.2(0.8)•	73.5(1.3)•
AMMSS	86.2(2.4)⊙	88.0(1.7)•	87.5(7.3)•	99.2(1.0)⊙	97.3(0.3)•	64.7(4.3)•	64.0(1.0)•	99.2(0.2)•	93.1(0.2)•
AMGL	84.9(2.7)•	85.3(2.0)•	92.1(2.6)•	90.2(2.0)•	97.3(0.3)•	63.9(4.3)•	63.6(1.0)•	97.7(0.3)•	90.5(0.3)•
MVL-IV	**87.9(2.0)**⊙	80.6(1.8)•	90.7(8.0)•	65.9(20.5)•	95.9(0.3)•	54.2(3.7)•	56.5(1.4)•	87.7(1.3)•	77.0(0.3)•
DMF	81.4(2.4)•	83.7(1.4)•	91.6(1.6)•	62.8(2.5)•	96.4(0.3)•	69.6(2.1)⊙	66.5(0.7)•	97.6(0.2)•	88.2(0.2)•
iMVWL	80.6(4.5)•	82.4(2.0)•	85.5(1.5)•	98.7(0.6)•	88.4(0.6)•	66.7(2.7)•	61.1(1.4)•	84.8(2.1)•	79.9(0.5)•
SLIM	86.0(2.1)⊙	82.9(1.7)•	89.3(0.9)•	98.0(0.7)•	94.3(0.8)•	67.9(2.9)•	68.0(0.5)•	97.8(0.2)•	93.5(0.3)•
AMSC	86.5(2.5)	**89.0(1.5)**	**98.9(0.6)**	**99.7(0.4)**	**97.6(0.2)**	**72.9(3.8)**	**68.8(0.9)**	**99.3(0.1)**	**93.7(0.2)**
Metric	F-score								
LP$_{vote}$	81.0(2.6)•	80.5(2.0)•	78.0(7.6)•	83.3(5.7)•	95.6(0.4)•	65.6(4.3)•	56.8(1.4)•	94.1(0.8)•	72.7(1.5)•
AMMSS	84.9(3.2)⊙	83.8(2.9)•	71.5(17.6)•	99.2(1.0)⊙	97.3(0.3)•	63.8(4.9)•	59.9(1.2)•	99.2(0.2)•	93.1(0.3)•
AMGL	82.8(3.6)•	79.4(2.8)•	85.7(5.7)•	90.2(2.0)•	97.3(0.3)•	62.9(4.9)•	59.5(1.2)•	97.7(0.3)•	90.5(0.3)•
MVL-IV	**86.7(2.6)**⊙	73.6(2.7)•	78.6(23.8)•	55.9(29.1)•	95.9(0.3)•	51.5(4.4)•	54.6(1.4)•	87.6(1.3)•	77.0(0.2)•
DMF	80.4(2.7)•	76.5(2.4)•	87.9(2.0)•	62.7(2.5)•	96.4(0.3)•	68.0(2.3)⊙	65.1(0.9)⊙	97.6(0.2)•	88.2(0.2)•
iMVWL	79.9(4.9)•	75.5(3.0)•	72.5(4.3)•	98.7(0.6)•	88.4(0.6)•	66.3(2.8)•	59.4(1.3)•	84.5(2.0)•	79.7(0.6)•
SLIM	84.7(2.5)⊙	78.0(2.4)•	80.6(2.0)•	98.0(0.7)•	94.4(0.8)•	66.5(3.4)⊙	65.1(0.5)⊙	97.7(0.2)•	93.5(0.3)•
AMSC	85.1(3.2)	**84.9(2.2)**	**98.3(1.0)**	**99.7(0.4)**	**97.6(0.2)**	**69.4(6.4)**	**65.9(1.3)**	**99.3(0.1)**	**93.7(0.2)**

windows (PIX), 47 Zernike moment (ZER) and 6 morphological (MOR) features.

(6) Indoor consists 67 indoor categories. Following [52], we select 5 categories (auditorium, buffet, classroom, cloister and elevator) with total 621 images. For each image, six visual features are extracted: 200 SIFT, 200 SURF, 680 PHOG, 256 LBP, 512 GIST and 32 wavelet texture (WT).

(7) Scene15 is composed of 4485 images belonging to 15 categories: highway, inside of cities, tall buildings, streets, suburb residence, forest, coast, mountain, open country, bedroom, kitchen, livingroom, office, industrial and store. The same six types of features with Indoor are extracted.

(8) KSA includes four subjects performing five actions [53]. Here, each action is regarded as a class. We select 2000 video frames from each action, forming a subset of 10,000 examples. We use four pose features, *i.e.*, 360 $f_{\mathrm{JJ_d}}$, 420 $f_{\mathrm{JL_d}}$, 65 $f_{\mathrm{LL_a}}$ and 10 $f_{\mathrm{PP_a}}$ [54] extracted by Ma et al [53].

(9) MNIST8M contains 8100,000 images for 0 to 9 handwritten digit classes [55]. From each class, we randomly select 2,000 examples, forming a subset MNIST with 20,000 examples. For each image, two features are extracted: 512-dim GIST and 32-dim WT.

B. Baselines

In the experiments, we use Adaptive Multi-Model Semi-Supervised classification [10] (AMMSS), Auto-weighted Multiple Graph Learning [11] (AMGL), matrix completion by Deep Matrix Factorization [17] (DMF), Incomplete Multiple Kernel K-means Algorithm with Mutual Kernel Completion [19] (MKKM-IK-MKC), Label Propagation [56] (LP), Multi-View Learning with Incomplete Views [21] (MVL-IV), Incomplete Multi-View Weak-Label Learning (iMVWL) [29] and Semi-supervised Multi-modal Learning with Incomplete Modalities [30] (SLIM) to generate baselines. The traditional multi-view classification methods focus on supervised [14], [15], [13] or semi-supervised [3] classification tasks. Moreover, they take full data representations as the input and can not process the multi-view data with incomplete features directly. Thus, they investigate different problems to ours and we have not compare with these four methods in our experiments.

- LP$_{vote}$: We first perform (11) on each view to obtain the classification results of present samples, then determine the label according to the vote of results.
- AMMSS$_{MC}$: We first concatenate data matrices of multiple views and fill the missing entries of them by DMF, and then implement AMMSS on the completed multi-view data to obtain the classification results.
- AMGL$_{MC}$: Similar to AMMSS$_{MC}$, we apply AMGL on the completed multi-view data.
- AMMSS$_{KC}$: We first construct partial graph matrices on each view, then complete these matrices by MKKM-IM-MKC and obtain the classification results by AMMSS.
- AMGL$_{KC}$: Similar to AMMKC$_{KC}$, we apply AMGL on the completed graph matrices.
- MVL-IV: We apply MVL-IV to learn a common representation matrix, then apply LP on it for classification.
- DMF: We apply DMF on the concatenated feature matrix of all views to learn a representation matrix, then apply LP on it to obtain the classification results.
- iMVWL: We apply it on incomplete multi-view data.
- SLIM: SLIM is a semi-classification method for incomplete multi-view data.

C. Experimental Setup

1) Data Processing: Since all these datasets are originally complete, to simulate the partial view setting, we randomly remove some view samples from each data. Concretely, for each $\mathbf{x}_i^{(v)}$, there is a probability to delete it. In the experiments, the probability ranges form 0% to 50% with a step 10%. And for each data $\mathbf{x}_i$, we ensure that it has at least one view sample $\mathbf{x}_i^{(v)}$ remaining. The probability can be regarded as the partial example ratio (PER) of the dataset. For each view, we normalize the data matrix, making all the values in the range $[0, 1]$. To mimic the real semi-supervised learning situation ($l \ll u$), 10% data points which are present on more views than the rest data points are selected to be assigned with labels. And we randomly select 10% data points from the rest data to form a validation set. The other 80% data points are unlabeled.

2) Parameter Determination: For our method, the adaptive parameter γ is tuned from 1.1 to 3.1 with a step 0.4. Following [11], [40], [41], [42], the parameter p is set to be 0.5.

TABLE II: Acc (%) comparisons on 9 datasets with different PERs. Standard deviation (%) is in the parentheses. The first highest score is in bold. Symbols '•/⊙/∘' denote that AMSC is better/tied/worse than the corresponding method by the paired t-test with confidence level 0.05, respectively. The win/tie/loss counts are summarized in the last row.

Dataset	PER	LP_{vote}	$AMMSS_{MC}$	$AMGL_{MC}$	$AMMSS_{KC}$	$AMGL_{KC}$	MVL-IV	DMF	iMVWL	SLIM	AMSC
MSRC-v1	10%	82.0(3.4)•	81.3(2.5)•	82.0(2.0)•	84.9(2.6)⊙	83.6(2.5)•	86.8(2.8)⊙	79.2(1.3)•	79.2(3.1)•	84.2(1.8)⊙	85.8(2.3)
	20%	78.9(3.4)•	82.8(2.6)⊙	82.6(1.9)⊙	82.5(1.5)•	80.8(1.6)•	83.3(3.1)⊙	76.9(2.6)•	77.8(4.5)•	83.1(2.2)⊙	**84.5(1.9)**
	30%	73.7(4.1)•	80.5(1.5)⊙	80.4(1.2)⊙	77.5(2.9)•	76.7(3.4)•	**82.9(3.3)**⊙	73.6(2.4)•	76.9(3.5)•	79.2(3.0)⊙	81.6(2.4)
	40%	69.4(3.6)•	75.6(2.8)•	75.9(2.8)•	74.4(4.1)•	73.8(2.9)•	**80.3(4.4)**⊙	67.9(3.8)•	73.7(2.9)•	73.5(2.9)•	**80.3(2.3)**
	50%	66.7(3.1)•	76.0(3.6)⊙	75.4(3.3)•	71.2(3.6)•	70.4(3.0)•	74.8(4.5)•	65.9(2.4)•	71.3(1.5)•	68.0(2.4)•	**78.4(2.3)**
Caltech7	10%	83.7(1.5)•	81.1(2.5)•	80.1(2.2)•	83.7(2.0)•	81.3(2.4)•	79.8(1.6)•	83.1(1.2)•	81.1(1.6)•	81.7(1.8)•	**87.9(1.6)**
	20%	82.4(1.9)•	79.8(1.4)•	78.7(1.9)•	80.3(3.2)•	78.4(3.0)•	78.6(1.4)•	81.8(0.7)•	80.7(2.2)•	80.4(1.0)•	**87.4(1.3)**
	30%	78.7(2.0)•	79.2(1.7)•	77.7(1.9)•	76.2(3.9)•	75.2(3.8)•	76.1(2.0)•	80.4(1.8)•	81.0(1.4)•	79.1(1.4)•	**84.9(1.7)**
	40%	75.7(1.3)•	77.3(2.3)•	75.2(2.4)•	72.4(2.5)•	72.0(2.5)•	74.3(3.1)•	78.3(1.3)•	78.3(1.9)•	75.9(1.9)•	**82.8(1.9)**
	50%	72.2(2.6)•	75.9(2.4)•	74.2(1.6)•	67.5(3.4)•	67.6(3.3)•	71.6(2.9)•	73.3(2.1)•	76.2(1.8)•	73.8(2.2)•	**80.6(2.3)**
Protein	10%	88.4(3.1)•	85.5(7.5)•	80.1(0.4)•	82.9(2.6)•	82.7(2.4)•	85.1(7.7)•	90.5(2.2)•	86.1(2.0)•	88.8(1.0)•	**98.7(0.6)**
	20%	85.7(2.1)•	83.3(5.4)•	81.7(2.2)•	82.2(5.1)•	80.6(0.5)•	88.2(6.4)•	90.1(2.4)•	85.5(1.7)•	88.3(2.0)•	**97.1(0.4)**
	30%	85.2(2.0)•	84.8(5.4)•	83.4(2.3)•	80.6(0.4)•	80.6(0.5)•	87.8(4.8)•	90.4(1.2)•	86.2(1.5)•	88.3(0.9)•	**95.5(1.1)**
	40%	87.0(2.2)•	89.6(4.6)•	84.2(3.1)•	87.9(5.9)•	87.8(4.8)•	84.8(5.1)•	89.6(0.7)•	85.7(0.9)•	87.5(1.2)•	**94.8(0.9)**
	50%	86.7(1.5)•	90.1(2.3)•	86.7(2.7)•	90.4(1.6)•	88.9(2.6)•	85.8(4.0)•	89.1(0.8)•	84.7(1.6)•	86.4(1.3)•	**92.7(1.1)**
NNSpt	10%	86.5(3.7)•	94.6(0.9)•	92.3(1.1)•	95.5(0.6)⊙	91.5(0.8)•	82.6(13.3)•	67.9(1.3)•	94.3(0.6)•	94.0(1.0)•	**95.7(0.6)**
	20%	82.7(3.9)•	92.1(1.7)⊙	89.7(1.2)•	92.3(1.2)⊙	89.1(2.1)•	68.4(19.4)•	70.8(1.9)•	89.7(0.8)•	91.5(1.5)⊙	**92.5(1.2)**
	30%	82.3(3.9)•	**90.1(0.6)**⊙	88.3(0.9)•	88.0(1.9)•	85.2(2.2)•	63.7(17.7)•	73.0(2.0)•	86.3(1.0)•	88.3(1.5)⊙	89.5(1.5)
	40%	82.3(1.8)•	**86.8(1.4)**⊙	85.6(1.4)⊙	84.7(1.7)•	82.6(2.3)•	69.6(16.9)•	75.3(2.4)•	83.7(1.5)•	85.5(2.1)⊙	86.4(2.0)
	50%	80.4(1.6)•	**84.8(2.1)**∘	84.2(1.5)∘	75.1(14.3)⊙	80.2(1.9)•	62.2(13.7)•	76.6(2.2)•	82.4(1.1)⊙	83.7(1.9)⊙	83.0(1.3)
Digits	10%	94.0(0.6)•	97.0(0.4)•	94.3(1.3)•	97.1(0.3)•	96.9(0.2)•	95.2(0.5)•	96.3(0.4)•	89.5(0.8)•	94.1(0.9)•	**97.3(0.3)**
	20%	93.6(0.5)•	**96.8(0.3)**⊙	94.3(0.7)•	96.3(0.5)•	95.8(0.4)•	94.2(0.9)•	95.6(0.4)•	88.8(0.9)•	93.9(0.8)•	**96.8(0.3)**
	30%	92.8(0.5)•	96.1(0.4)⊙	94.6(0.7)•	94.9(0.6)•	94.4(0.8)•	92.8(0.3)•	95.1(0.6)•	88.3(1.5)•	92.3(1.1)•	**96.2(0.3)**
	40%	89.2(0.7)•	94.7(0.5)•	93.8(0.6)•	91.2(1.3)•	90.0(1.4)•	90.5(0.9)•	93.6(0.5)•	87.4(1.0)•	89.9(1.0)•	**95.1(0.1)**
	50%	87.6(0.9)•	92.4(0.6)•	92.2(0.3)•	87.4(1.7)•	86.0(2.3)•	87.9(0.7)•	91.1(0.5)•	84.7(1.3)•	86.1(1.1)•	**92.8(0.4)**
Indoor	10%	67.7(2.6)•	63.0(3.5)•	62.7(3.3)•	63.1(4.9)•	62.7(5.0)•	52.0(4.2)•	65.6(1.6)•	66.2(2.6)•	68.0(1.6)•	**73.1(1.9)**
	20%	66.1(2.7)•	56.4(4.6)•	56.6(5.0)•	61.9(3.9)•	61.5(4.3)•	48.4(3.3)•	63.5(2.4)•	64.7(2.5)•	65.3(2.7)•	**71.1(2.9)**
	30%	64.6(2.1)•	57.0(5.3)•	57.2(5.2)•	61.9(2.9)•	61.4(3.1)•	46.1(4.1)•	59.8(1.8)•	63.0(2.0)•	65.3(3.1)•	**72.0(1.9)**
	40%	64.1(2.7)•	50.9(5.7)•	51.8(5.7)•	62.6(4.8)•	62.5(4.5)•	41.7(4.2)•	56.3(2.3)•	60.0(2.6)•	61.8(1.7)•	**70.6(1.7)**
	50%	59.8(2.5)•	50.7(4.3)•	50.6(4.0)•	54.2(9.3)•	55.3(9.8)•	39.4(3.5)•	53.1(2.5)•	56.4(2.1)•	55.9(2.4)•	**66.6(2.1)**
Scene15	10%	59.5(1.0)•	63.3(0.6)•	63.0(0.6)•	63.0(1.5)•	62.7(1.6)•	55.4(1.4)•	65.9(0.7)•	59.4(1.7)•	67.4(0.7)⊙	**67.8(1.1)**
	20%	56.4(0.8)•	62.3(0.8)•	62.0(0.8)•	60.5(1.0)•	60.2(1.0)•	52.6(1.9)•	64.9(0.9)⊙	58.4(1.5)•	64.9(0.9)•	**65.9(0.9)**
	30%	53.7(1.0)•	60.1(1.0)•	59.8(1.0)•	58.1(1.1)•	58.0(0.9)•	50.6(1.4)•	62.9(1.0)⊙	57.0(1.5)•	**63.0(0.7)**⊙	**63.0(0.4)**
	40%	51.0(0.7)•	58.4(1.6)•	58.2(1.6)•	56.4(1.4)•	56.1(1.3)•	49.4(1.3)•	**60.9(1.3)**⊙	54.1(1.2)•	59.6(1.0)•	60.8(0.7)
	50%	46.7(0.9)•	56.9(0.9)⊙	56.7(0.8)⊙	51.4(2.2)•	49.7(7.2)•	42.4(13.1)•	**57.2(0.7)**∘	51.3(1.2)•	55.7(0.7)⊙	56.4(0.7)
KSA	10%	93.4(0.6)•	98.3(0.2)•	97.0(0.3)•	96.9(0.6)•	94.1(0.6)•	85.2(1.6)•	97.1(0.2)•	85.2(1.9)•	97.1(0.5)•	**99.2(0.1)**
	20%	93.8(0.4)•	96.8(0.2)•	95.7(0.4)•	92.6(0.9)•	90.5(1.0)•	82.7(1.5)•	95.9(0.3)•	84.4(1.6)•	96.3(0.4)•	**98.6(0.1)**
	30%	92.3(0.5)•	94.8(0.4)•	94.0(0.3)•	86.8(1.3)•	85.3(1.3)•	78.9(1.5)•	93.5(0.3)•	82.6(1.7)•	95.1(0.2)•	**97.8(0.2)**
	40%	91.5(0.6)•	92.3(0.5)•	92.1(0.4)•	80.6(1.1)•	79.4(1.0)•	74.0(1.5)•	91.3(0.8)•	81.0(1.5)•	93.3(0.4)•	**96.2(0.1)**
	50%	90.8(0.7)•	89.4(0.4)•	89.6(0.3)•	75.3(0.5)•	74.9(0.6)•	71.5(0.9)•	87.5(0.7)•	77.0(1.4)•	91.0(0.3)•	**94.7(0.1)**
MNIST	10%	74.0(0.7)•	89.1(0.2)•	87.4(0.3)•	88.3(0.2)•	86.7(0.3)•	69.2(6.8)•	79.2(0.3)•	76.6(0.9)•	88.7(0.2)•	**90.1(0.2)**
	20%	74.5(0.7)•	85.2(0.5)•	84.1(0.6)•	84.1(0.5)•	82.9(0.6)•	64.4(3.7)•	72.7(0.4)•	72.3(1.2)•	84.2(0.3)•	**86.9(0.4)**
	30%	75.3(0.4)•	81.7(0.4)•	81.1(0.4)•	80.2(0.5)•	79.5(0.5)•	63.3(3.7)•	67.2(0.3)•	68.7(1.5)•	80.2(0.4)•	**83.9(0.3)**
	40%	75.3(0.4)•	78.0(0.4)•	77.9(0.3)•	76.7(0.4)•	76.2(0.4)•	58.9(3.2)•	62.3(0.4)•	64.7(0.7)•	76.7(0.3)•	**81.3(0.3)**
	50%	75.3(0.3)•	74.8(0.2)•	75.0(0.3)•	73.7(0.5)•	73.6(0.6)•	59.6(3.9)•	58.1(0.6)•	60.9(0.8)•	73.8(0.2)•	**79.0(0.3)**
win/tie/lose		45/0/0	35/9/1	40/4/1	41/4/0	45/0/0	41/4/0	41/3/1	44/1/0	35/10/0	-

We set $m^{(v)} = n^{(v)}$ on MSRC-v1, Caltech7 and Protein since $d^{(v)} \gg n^{(v)}$. And we set $m^{(v)} = 200$ on NNspt, Digits and Indoor, and $m^{(v)} = 500$ on Scene15, KSA and MNIST to further accelerate the algorithm. For the compared methods, we download the source codes from the authors' websites and determine the searching ranges of the parameters according to their papers. In the experiment, the partial graph matrices are constructed according to Eq. (8) and the complete graph matrices are constructed by [39]. The neighbor number n_k is fixed to be 5 for all experiments. All hyper-parameters are tuned by grid search on the validation set, and the results of using the best tuned parameters are recorded.

3) Evaluation Metric: The classification performance is evaluated in terms of classification accuracy (Acc) and F-score. For a fair comparison, on every multi-view dataset, each time we create incomplete datasets with different PERs of missing examples and perform 10 independent times. The average result with standard deviation (STD) is reported.

D. Classification Results Comparison

When PER$=0$, AMMSS and AMGL can be applied on data directly. Thus, we report Acc and F-score comparisons of seven methods on nine datasets with PER$=0$ in Table I. Table II and Table III show Acc and F-score comparisons of all nine compared methods with varying PER, respectively.

From Table I, II and III, we have the following observations: (1) The performance of each method tends to be worse as the increase of partial example ratio in most cases, which is consistent with intuition. (2) LP_{vote} uses the simplest way to integrate the classification results of multiple models. When data are complete, AMMSS and AMGL have achieved better or comparable performance than LP_{vote} on all datasets except Indoor. However, in incomplete multi-view setting, these two methods facilitated with two kind of completion methods have achieved significantly worse performance than LP_{vote} on datasets Caltech7 and Protein in some cases, which illustrates completion is not always an advisable manner to deal with incomplete multi-view data. (3) The results of MVL-IV are lower than DMF with a big gap on NNSpt, Indoor, KSA and MNIST. The possible reason is that MVL-IV learns the common representation matrix linearly, which cannot well disclose non-linear structure. (4) $AMMSC_{MC}$ and $AMGL_{MC}$ outperform $AMMSC_{KC}$ and $AMGL_{KC}$ in most cases. It is probably because graph completion may destroy the local structure. (5) On incomplete datasets, Slim achieved better or comparable results than other methods expect AMSC in

TABLE III: F-score (%) comparisons on 9 datasets with different PERs. (See the title of Table II for more information).

Dataset	PER	LP_{vote}	$AMMSS_{MC}$	$AMGL_{MC}$	$AMMSS_{KC}$	$AMGL_{KC}$	MVL-IV	DMF	iMVWL	SLIM	AMSC
MSRC-v1	10%	80.8(3.8)•	79.7(3.1)•	80.3(2.7)•	82.5(3.6)⊙	80.6(3.5)•	**85.5(3.5)**⊙	78.7(1.4)•	78.1(3.0)•	83.0(2.2)⊙	84.0(3.1)
	20%	77.7(3.7)•	81.1(3.7)⊙	80.7(2.9)⊙	79.1(1.8)•	76.9(1.9)•	81.4(3.9)⊙	76.3(2.4)•	76.3(5.4)•	81.8(3.1)⊙	**83.0(2.3)**
	30%	72.6(4.0)•	79.3(1.5)⊙	79.1(1.1)⊙	73.3(3.0)•	72.3(3.8)•	**81.6(4.0**⊙	73.0(3.2)•	75.8(3.8)⊙	78.3(3.1)⊙	79.6(3.1)
	40%	68.3(4.0)•	73.6(3.2)•	73.8(2.9)•	70.4(5.2)•	69.6(3.7)•	**79.3(4.7)**⊙	67.3(3.2)•	72.4(3.5)•	72.9(3.1)•	78.7(2.4)
	50%	66.1(3.5)•	74.4(4.4)⊙	73.8(3.9)⊙	67.7(4.0)•	66.8(3.2)•	73.1(5.0)⊙	65.4(2.6)•	70.2(1.5)•	67.2(2.7)•	**76.3(3.3)**
Caltech7	10%	78.7(2.3)•	74.9(2.6)•	72.8(3.2)•	76.6(3.4)•	73.7(3.9)•	72.2(2.2)•	76.5(2.3)•	74.3(2.3)•	76.6(2.2)•	**83.6(2.5)**
	20%	76.7(2.4)•	71.0(2.9)•	69.4(3.1)•	71.8(5.2)•	68.9(4.4)•	72.3(1.1)•	74.8(1.7)•	73.6(3.4)•	74.5(1.5)•	**82.7(1.9)**
	30%	72.1(3.4)•	71.6(2.7)•	68.1(3.2)•	66.0(5.9)•	64.5(6.2)•	69.1(3.2)•	74.5(2.4)•	75.1(1.8)•	72.7(2.8)•	**79.5(2.4)**
	40%	69.3(2.1)•	69.3(4.9)•	66.1(4.7)•	60.4(4.0)•	59.5(3.7)•	67.1(3.8)•	72.1(1.2)•	72.5(2.9)•	70.2(2.0)•	**76.9(3.0)**
	50%	64.6(3.2)•	67.7(3.2)•	64.2(2.8)•	52.5(4.2)•	52.2(4.5)•	63.6(3.8)•	66.1(2.8)•	69.6(2.9)•	67.1(3.7)•	**73.8(3.6)**
Protein	10%	77.3(9.0)•	65.5(21.0)•	48.7(2.1)•	59.8(8.7)•	59.3(7.9)•	62.8(24.1)•	85.1(4.0)•	73.6(4.6)•	80.1(2.2)•	**98.1(0.9)**
	20%	70.1(6.0)•	59.8(16.1)•	55.2(8.8)•	55.7(13.9)•	51.1(2.3)•	74.9(21.3)•	85.0(3.6)•	72.0(4.9)•	79.3(4.6)•	**95.5(0.7)**
	30%	68.4(6.1)•	66.0(16.1)•	61.9(8.0)•	51.0(1.9)•	51.1(2.2)•	76.1(17.3)•	85.0(2.2)•	73.4(4.4)•	79.6(1.6)•	**92.8(2.0)**
	40%	74.3(6.5)•	80.4(13.4)•	64.4(10.2)•	73.5(16.6)•	74.6(13.8)•	66.5(19.5)•	83.8(1.4)•	72.3(2.3)•	78.3(2.2)•	**91.6(1.6)**
	50%	73.8(4.0)•	83.0(4.4)•	73.1(8.6)•	82.6(4.1)•	79.0(6.6)•	72.0(15.4)•	82.2(1.4)•	69.4(4.3)•	77.0(2.1)•	**87.8(2.0)**
NNSpt	10%	86.4(3.8)•	94.6(0.9)•	92.3(1.1)•	95.5(0.6)⊙	91.5(0.9)•	81.0(18.0)•	67.8(1.2)•	94.3(0.6)•	94.0(1.0)•	**95.7(0.6)**
	20%	82.5(4.2)•	92.1(1.8)⊙	89.7(1.2)•	92.3(1.2)⊙	89.1(2.1)•	60.0(28.2)•	70.7(2.0)•	89.7(0.8)•	91.4(1.6)⊙	**92.5(1.2)**
	30%	82.2(4.1)•	**90.1(0.6)**⊙	88.3(0.9)•	87.9(1.9)•	85.2(2.2)•	53.7(26.3)•	73.0(2.0)•	86.3(1.0)•	88.1(1.6)⊙	89.5(1.5)
	40%	82.3(1.8)•	**86.8(1.4)**⊙	85.6(1.4)⊙	84.6(1.7)•	82.5(2.4)•	62.9(25.5)•	75.2(2.4)•	83.7(1.5)•	85.3(2.3)⊙	86.4(2.0)
	50%	80.4(1.6)•	**84.7(2.2)**○	84.2(1.5)○	74.8(15.2)⊙	80.1(2.0)•	53.8(22.1)•	76.6(2.3)•	82.4(1.1)⊙	83.4(2.1)⊙	83.0(1.3)
Digits	10%	93.9(0.6)•	97.0(0.4)•	94.3(1.4)•	97.0(0.3)•	96.9(0.2)•	95.2(0.5)•	96.3(0.3)•	89.4(0.9)•	94.1(0.9)•	**97.3(0.3)**
	20%	93.6(0.6)•	**96.8(0.3)**⊙	94.3(0.7)•	96.2(0.5)•	95.8(0.4)•	94.2(0.9)•	95.6(0.4)•	88.8(0.9)•	93.9(0.8)•	**96.8(0.3)**
	30%	92.7(0.5)•	96.1(0.4)⊙	94.6(0.7)•	94.9(0.6)•	94.3(0.8)•	92.9(0.4)•	95.1(0.6)•	88.3(1.5)•	92.3(1.1)•	**96.2(0.3)**
	40%	89.1(0.7)•	94.7(0.5)•	93.8(0.6)•	91.1(1.4)•	89.8(1.5)•	90.6(0.9)•	93.6(0.5)•	87.4(1.1)•	89.8(1.0)•	**95.1(0.1)**
	50%	87.4(0.9)•	92.4(0.6)⊙	92.1(0.4)•	87.0(1.8)•	85.5(2.6)•	87.9(0.7)•	91.2(0.4)•	84.7(1.2)•	86.1(1.1)•	**92.8(0.4)**
Indoor	10%	66.1(3.2)•	61.8(4.0)•	61.4(3.6)•	61.6(6.0)•	61.1(6.0)•	49.7(4.4)•	64.6(1.8)•	66.0(2.6)•	66.8(1.5)•	**71.4(2.4)**
	20%	64.3(3.3)•	53.5(5.6)•	53.7(6.1)•	59.9(4.7)•	59.6(5.1)•	45.1(3.5)•	62.3(2.4)•	64.0(2.7)•	64.1(2.9)•	**68.8(4.1)**
	30%	63.2(2.2)•	54.2(6.9)•	54.5(6.8)•	59.3(4.4)•	58.7(4.5)•	43.5(4.3)•	58.4(1.6)•	62.4(2.2)•	64.1(3.3)•	**70.7(2.2)**
	40%	62.6(2.6)•	46.1(7.7)•	47.5(7.3)•	60.0(6.3)•	59.8(6.0)•	39.1(3.7)•	55.3(2.7)•	59.5(2.7)•	60.6(1.9)•	**69.1(1.9)**
	50%	58.4(2.8)•	45.1(6.0)•	45.0(5.5)•	49.6(9.2)•	50.6(9.7)•	35.7(3.5)•	52.0(2.6)•	55.3(2.1)•	54.5(2.5)•	**65.1(2.5)**
Scene15	10%	55.2(1.2)•	59.3(1.1)•	58.9(1.1)•	59.0(2.0)•	58.6(2.0)•	53.4(1.4)•	64.2(0.8)•	57.3(1.8)•	64.9(0.8)⊙	**65.4(1.2)**
	20%	51.7(1.2)•	58.0(1.1)•	57.7(1.0)•	55.6(1.5)•	55.2(1.5)•	50.7(1.7)•	63.3(1.0)⊙	56.6(1.7)•	62.1(0.9)•	**63.1(1.3)**
	30%	49.1(1.4)•	55.5(1.6)•	55.2(1.5)•	53.1(1.4)•	52.9(1.5)•	48.8(1.5)•	**61.1(1.1)**⊙	55.0(1.6)•	60.3(0.6)⊙	60.1(0.7)
	40%	46.4(0.9)•	54.0(2.5)•	53.8(2.5)•	50.6(2.4)•	50.1(2.2)•	47.4(1.2)•	**59.1(1.5)**○	51.9(1.3)•	56.9(1.3)•	57.7(1.0)
	50%	42.3(0.9)•	52.5(1.0)⊙	52.3(1.0)•	45.7(2.5)•	44.4(5.7)•	40.0(13.9)•	**55.1(0.8)**○	48.6(1.3)•	53.1(0.8)⊙	53.4(1.0)
KSA	10%	93.4(0.6)•	98.3(0.2)•	96.9(0.3)•	96.9(0.6)•	94.1(0.6)•	85.2(1.6)•	97.1(0.2)•	84.9(1.9)•	97.0(0.5)•	**99.2(0.1)**
	20%	93.8(0.5)•	96.7(0.2)•	95.7(0.4)•	92.5(0.9)•	90.5(1.0)•	82.7(1.5)•	95.9(0.3)•	84.1(1.6)•	96.3(0.4)•	**98.6(0.1)**
	30%	92.2(0.5)•	94.8(0.4)•	94.0(0.4)•	86.8(1.3)•	85.2(1.3)•	78.9(1.5)•	93.5(0.3)•	82.4(1.8)•	95.1(0.2)•	**97.8(0.2)**
	40%	91.5(0.6)•	92.2(0.5)•	92.1(0.4)•	80.4(1.0)•	79.2(0.9)•	74.1(1.4)•	91.3(0.8)•	80.8(1.5)•	93.3(0.4)•	**96.2(0.1)**
	50%	90.8(0.7)•	89.3(0.4)•	89.5(0.3)•	75.0(0.5)•	74.5(0.7)•	71.5(0.9)•	87.6(0.7)•	76.7(1.5)•	90.9(0.3)•	**94.6(0.1)**
MNIST	10%	73.5(0.7)•	89.1(0.2)•	87.4(0.3)•	88.3(0.2)•	86.7(0.3)•	69.0(6.9)•	79.8(0.3)•	76.4(0.9)•	88.7(0.2)•	**90.2(0.2)**
	20%	74.2(0.7)•	85.2(0.5)•	84.1(0.6)•	84.2(0.5)•	82.9(0.6)•	64.1(3.8)•	74.1(0.4)•	72.2(1.2)•	84.2(0.3)•	**87.0(0.4)**
	30%	75.1(0.5)•	81.8(0.4)•	81.1(0.4)•	80.2(0.5)•	79.5(0.5)•	62.8(3.9)•	69.3(0.5)•	68.4(1.6)•	80.2(0.4)•	**83.9(0.3)**
	40%	75.2(0.4)•	78.0(0.4)•	77.9(0.3)•	76.6(0.4)•	76.2(0.4)•	58.2(3.6)•	64.6(0.5)•	64.2(0.9)•	76.5(0.3)•	**81.3(0.3)**
	50%	75.3(0.3)•	74.7(0.3)•	74.9(0.3)•	73.6(0.5)•	73.5(0.6)•	59.0(4.2)•	61.5(1.0)•	60.5(1.0)•	73.6(0.2)•	**79.0(0.3)**
win/tie/lose		45/0/0	35/9/1	40/4/1	41/4/0	45/0/0	40/5/0	41/2/2	43/2/0	35/10/0	-

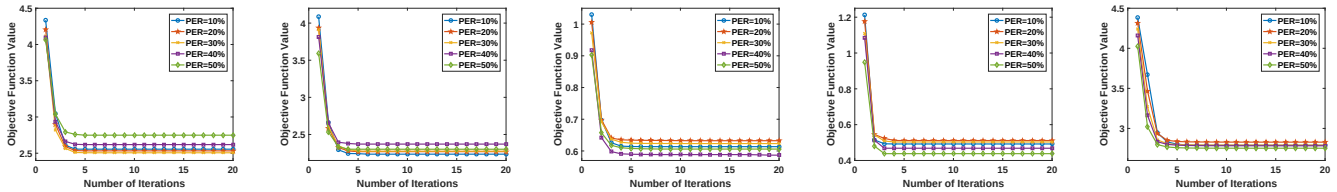
TABLE IV: Average runtime comparison (seconds) on 9 datasets with PER$= 50\%$.

Dataset	MSRC-v1	Caltech7	Protein	NNSpt	Digits	Indoor	Scene15	KSA	MNIST
LP_{vote}	0.1034	0.1156	0.3789	0.1613	0.8501	0.1650	6.7755	21.8805	134.7194
$AMMSS_{MC}$	20.8244	33.3290	112.9571	22.3328	46.2888	27.4227	282.5742	1054.0155	19593.6493
$AMGL_{MC}$	20.7244	33.0138	112.2643	21.7133	39.8355	27.0254	237.4319	291.6899	986.9956
$AMMSS_{KC}$	1.5281	2.8514	2.2334	2.6695	81.4441	3.9942	1192.0755	3032.5883	14240.0511
$AMGL_{KC}$	1.2030	2.4553	1.3083	1.6413	71.8105	3.4752	976.4958	2684.2480	6567.0985
MVL-IV	2.2217	2.8878	28.4404	0.2438	5.7611	3.6038	24.4841	129.9938	421.8936
DMF	20.7244	32.7914	111.6507	21.4363	37.6756	26.7435	224.2482	195.3210	644.0667
iMVWL	4.3500	5.5158	25.8585	5.0287	7.8614	5.2581	30.4154	24.6500	45.0522
Slim	7.1595	8.5050	11.5334	3.7400	33.7672	6.2025	680.3511	3321.4526	51564.3778
AMSC	0.1307	0.2306	0.4460	0.2019	0.9194	0.2892	8.5192	10.0985	15.7300

most cases. (6) When PER$= 0$, AMSC has achieved better or comparable performance than other methods on all datasets, indicating that AMSC is also an effective method for traditional semi-supervised classification task. When PER varies from 10% to 50%, AMSC outperforms other methods in terms of both Acc and F-score in most cases, and the advantages seem more obvious than PER=0. Because AMSC is directly designed for incomplete multi-view data, which takes the contribution importance of different instances and views into consideration, and establishes a reasonable interaction between the learned common and view-specific label matrices.

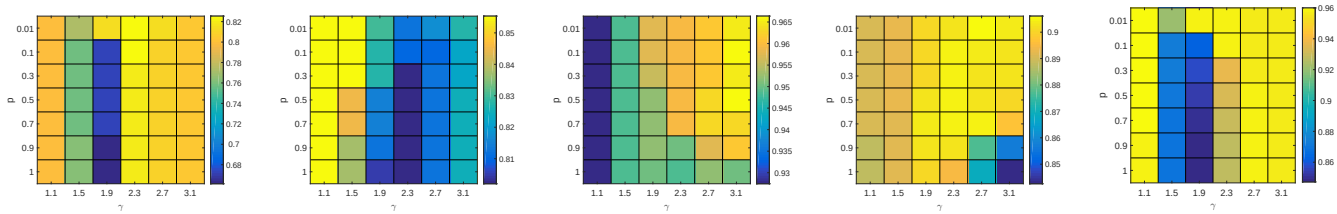
E. Time Comparison and Convergence Analysis

To demonstrate the efficiency of AMSC to deal with incomplete multi-view data, we report the runtime comparisons on 9 datasets with PER$= 50\%$ in Table IV. From Table IV, we have the following observations: (1) LP_{vote} takes the least time on all datasets except KSA and MNIST, because optimization has cubic complexity w.r.t. the present data size without iterations. (2) Since AMSC has linear complexity w.r.t both present data size and dimensionality, it spends the least time on KSA and MNIST, and the second least time on other datasets. (3) On datasets MSRC-v1, Caltech7, Protein, NNSpt and Indoor, where dimensionality is larger than data size, DMF, $AMMSS_{MC}$ and $AMGL_{MC}$ spend more time than other methods because matrix completion has quadratic complexity w.r.t the dimensionality and needs iterations. SLIM consumes more time than $AMMSS_{KC}$ and $AMGL_{KC}$ because it has cubic complexity w.r.t the dimensionality of each view. (4) On datasets Digits, Scene15, KSA and MNIST, where the data size



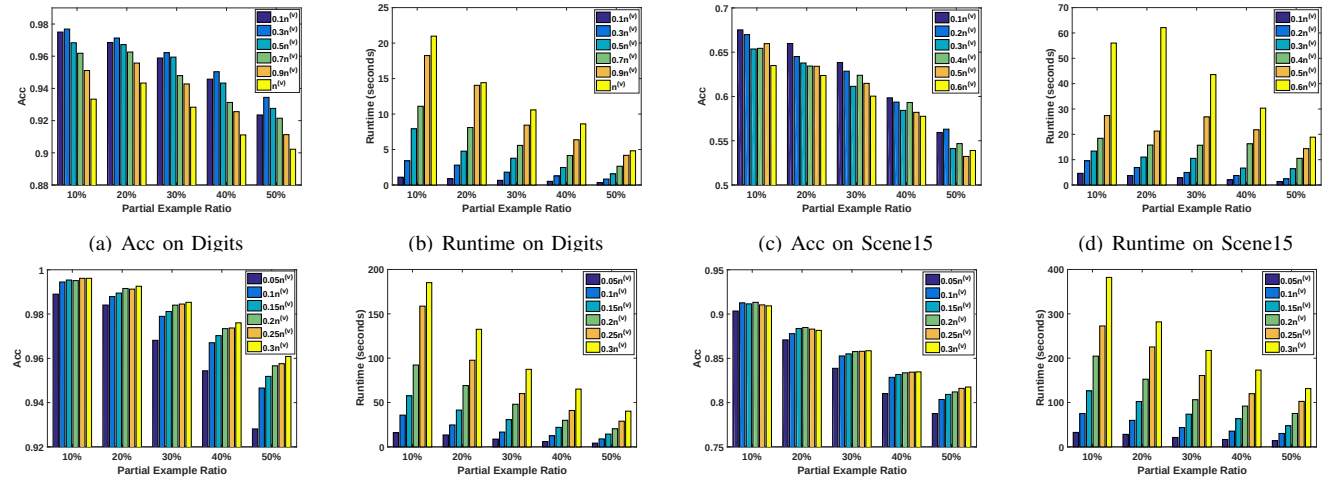
(a) MSRC-v1 (b) Caltech7 (c) Protein (d) NNSpt (e) Digits

Fig. 1: The convergence curves of AMSC in Eq. (17) on 5 datasets with different PERs.



(a) MSRC-v1 (b) Caltech7 (c) Protein (d) NNSpt (e) Digits

Fig. 2: Sensitivity analysis of Acc on parameters p and γ with PER$=30\%$.



(a) Acc on Digits (b) Runtime on Digits (c) Acc on Scene15 (d) Runtime on Scene15

(e) Acc on KSA (f) Runtime on KSA (g) Acc on MNIST (h) Runtime on MNIST

Fig. 3: Sensitivity analysis on $m^{(v)}$ with different PERs.

is much larger than dimensionality, AMMSS$_{KC}$, AMGL$_{KC}$ and SLIM take more time because they have high complexity w.r.t data size and need iterations. MVL-IV spends third least time than other, because it has linear complexity w.r.t data size on the representation learning.

In order to verify the convergence behavior of Alg. 1, we plot the corresponding convergence curves of the objective of the problem (17) on datasets MSRC-v1, Caltech7, Protein, NNSpt and Digits. From Fig. 1, It can be observed that the objective function value monotonically decreases as the iteration round increases and converges to a fixed value. Additionally, the algorithm converges within a few iterations over all datasets, indicating Alg. 1 has fine convergence speed.

F. Parameter Study

We first study the influence of hyper-parameters p and γ on the performance of AMSC. p is tuned in the range of $\{0.01, 0.1, 0.3, 0.5, 0.7, 0.9, 1\}$ while γ is varied from $\{1.1, 1.5, 1.9, 2.3, 2.7, 3.1\}$. On each data, we set the same $m^{(v)}$ as the subsection of experimental setup, and PER is fixed as 30%. Since F-score has similar tendency with Acc, Fig. 2 shows Acc results with varying parameters p and γ on five datasets, i.e., MSRC-v1, Caltech7, Protein, NNspt and Digits. As Fig. 2 shown that five datasets have different optimal parameters. On datasets MSRC-v1 and Digits, the optimal combination of parameters is mainly determined by γ, whereas on datasets Caltech7, Protein and NNSpt, the performance is affected by both γ and p. Therefore, how to identify the optimal parameters is data dependent. Besides, by fixing $p = 0.5$ and varying γ, AMSC can achieve satisfactory results on all five datasets, which demonstrates that empirically setting p as 0.5 is reasonable.

Then we study the influence of the number of anchors $m^{(v)}$ on AMSC. Since anchor-based strategy is usually effective and efficient when data size is significantly larger than dimensionality, the experiments are conducted on Digits, Scene15, KSA and MINIST. p is fixed as 0.5. On Digits and Scene15, we vary $m^{(v)}$ from $\{0.1n^{(v)}, 0.3n^{(v)}, 0.5n^{(v)}, 0.7n^{(v)}, 0.9n^{(v)}, n^{(v)}\}$ and $\{0.1n^{(v)}, 0.2n^{(v)}, 0.3n^{(v)}, 0.4n^{(v)}, 0.5n^{(v)}, 0.6n^{(v)}\}$, respectively. On KSA and MINIST, we vary $m^{(v)}$ from $\{0.05n^{(v)}, 0.1n^{(v)}, 0.15n^{(v)}, 0.2n^{(v)}, 0.25n^{(v)}, 0.3n^{(v)}\}$. Fig. 3 show Acc and runtime comparisons of varying $m^{(v)}$ with different PERs on four datasets. As shown in Fig. 3, with the increase of the number of anchors, on Digits and Scene15,

TABLE V: Details of 6 incomplete multi-view datasets (present data size (dimensionality)).

View	BBC2	BBC3	BBC4	BBCSport2	BBCSport3	BBCSport4
1	2125 (6838)	1828 (5470)	1543 (4659)	644 (3183)	519 (2582)	400 (1991)
2	2112 (6790)	1832 (5549)	1524 (4633)	637 (3203)	531 (2544)	410 (2063)
3	-	1845 (5483)	1574 (4665)	-	513 (2465)	437 (2113)
4	-	-	1549 (4684)	-	-	432 (2158)
Size	2225			737		
Classes	5			5		

TABLE VI: Acc (%) comparisons on 6 datasets with different LERs. (See the title of Table II for more information).

Dataset	LER	LP$_{vote}$	AMMSS$_{MC}$	AMGL$_{MC}$	AMMSS$_{KC}$	AMGL$_{KC}$	MVL-IV	DMF	iMVWL	SLIM	AMSC
BBC2	5%	91.5(1.0)•	91.5(0.8)•	91.7(0.9)•	91.3(1.1)•	91.4(1.0)•	92.6(0.8)•	93.0(0.7)•	90.9(1.5)•	90.8(1.8)•	**93.8(1.0)**
	10%	92.7(0.7)•	93.2(0.7)•	93.3(0.8)•	93.0(0.9)•	93.1(1.0)•	92.9(0.4)•	93.6(0.4)•	91.7(1.4)•	94.2(0.6)⊙	**94.6(0.7)**
	15%	93.1(0.5)•	94.0(0.5)•	94.1(0.4)•	93.7(0.6)•	93.8(0.5)•	93.2(0.2)•	93.8(0.5)•	92.3(0.9)•	94.8(0.4)⊙	**95.1(0.5)**
	20%	93.3(0.5)•	94.2(0.4)•	94.1(0.5)•	94.1(0.6)•	94.2(0.6)•	93.2(0.5)•	94.1(0.4)•	91.8(1.7)•	**95.5(0.6)**⊙	95.3(0.3)
BBC3	5%	92.0(0.8)•	93.3(0.6)•	93.3(0.5)•	84.2(1.3)•	84.2(1.3)•	92.0(0.9)•	91.5(1.1)•	90.2(0.5)•	87.4(3.7)•	**94.0(0.5)**
	10%	93.0(0.4)•	94.3(0.4)•	94.3(0.4)•	89.9(0.9)•	89.7(1.0)•	92.8(0.3)•	92.3(0.6)•	90.8(0.8)•	92.1(1.0)•	**94.7(0.4)**
	15%	93.1(0.6)•	94.5(0.4)•	94.5(0.4)•	91.7(0.4)•	91.4(0.4)•	93.1(0.4)•	92.9(0.4)•	91.1(0.9)•	94.0(1.0)•	**95.1(0.5)**
	20%	93.9(0.5)•	94.9(0.4)•	94.9(0.4)•	92.5(0.7)•	92.3(0.7)•	92.9(0.3)•	93.2(0.4)•	91.1(1.1)•	94.4(0.5)•	**95.5(0.5)**
BBC4	5%	91.2(0.5)•	92.8(0.5)•	92.8(0.5)•	74.0(2.7)•	74.3(2.7)•	93.3(0.5)•	91.2(0.7)•	86.4(2.3)•	80.5(4.1)•	**93.8(0.5)**
	10%	92.2(0.4)•	93.7(0.4)•	93.7(0.4)•	81.0(0.7)•	81.2(0.7)•	93.7(0.5)•	92.3(0.6)•	87.8(1.4)•	87.1(1.8)•	**94.6(0.3)**
	15%	92.7(0.5)•	94.0(0.4)•	94.0(0.4)•	84.5(1.2)•	84.5(1.3)•	93.6(0.3)•	92.8(0.6)•	88.6(1.5)•	91.4(1.1)•	**94.9(0.3)**
	20%	92.9(0.4)•	94.0(0.3)•	94.0(0.3)•	86.7(0.5)•	86.7(0.5)•	93.7(0.4)•	93.0(0.4)•	87.4(2.3)•	93.0(0.6)•	**95.0(0.2)**
BBCSport2	5%	86.6(3.6)•	72.5(5.3)•	75.7(5.2)•	79.8(4.1)•	81.4(4.5)•	86.5(2.1)•	91.9(1.7)•	83.8(2.5)•	75.7(3.3)•	**95.4(0.6)**
	10%	92.0(0.8)•	89.5(1.4)•	90.2(1.3)•	86.8(1.7)•	88.0(1.9)•	88.2(1.3)•	92.3(0.5)•	86.6(2.0)•	80.2(2.0)•	**96.1(0.4)**
	15%	92.8(0.8)•	93.1(0.7)•	93.4(0.6)•	91.3(0.9)•	91.9(0.7)•	88.4(0.8)•	92.5(0.6)•	86.7(2.3)•	83.2(1.0)•	**96.3(0.2)**
	20%	94.3(0.5)•	94.4(0.7)•	94.5(0.7)•	92.9(0.7)•	93.3(0.9)•	89.4(0.9)•	92.9(0.4)•	85.4(2.1)•	86.3(1.3)•	**96.7(0.3)**
BBCSport3	5%	84.1(1.8)•	78.7(3.5)•	79.3(3.5)•	62.4(2.0)•	62.6(1.9)•	88.5(0.7)•	90.7(0.9)•	74.9(3.9)•	72.2(2.3)•	**93.0(0.7)**
	10%	89.6(1.2)•	88.2(1.4)•	88.5(1.4)•	69.6(2.5)•	70.0(2.5)•	89.0(0.7)•	90.5(0.9)•	78.9(3.6)•	78.9(1.7)•	**93.9(1.0)**
	15%	91.4(1.2)•	91.1(1.0)•	91.1(1.0)•	76.4(1.1)•	76.8(1.1)•	89.1(0.4)•	90.8(0.8)•	79.1(2.5)•	81.5(1.8)•	**94.7(0.7)**
	20%	91.3(0.7)•	91.9(0.6)•	91.9(0.5)•	79.8(0.6)•	80.2(0.6)•	88.8(0.5)•	91.3(1.0)•	78.0(3.1)•	84.0(1.0)•	**94.9(0.4)**
BBCSport4	5%	82.7(2.4)•	84.1(2.6)•	84.3(2.5)•	51.6(2.6)•	52.3(2.7)•	87.1(0.6)•	90.6(0.9)•	67.6(4.9)•	71.7(1.8)•	**91.9(0.6)**
	10%	87.9(0.5)•	90.6(0.7)•	90.5(0.7)•	67.0(1.2)•	67.3(1.2)•	88.7(0.2)•	91.3(0.5)•	72.0(3.9)•	75.9(1.2)•	**92.7(0.6)**
	15%	90.0(0.7)•	92.0(0.6)•	91.9(0.5)•	75.7(0.9)•	75.9(0.7)•	88.9(0.5)•	91.1(0.6)•	73.5(3.0)•	79.3(1.4)•	**93.8(0.4)**
	20%	91.3(0.6)•	92.0(0.3)•	91.9(0.3)•	78.2(0.4)•	78.5(0.4)•	88.6(0.4)•	90.4(0.4)•	73.6(2.9)•	81.5(1.0)•	**94.3(0.4)**
win/tie/lose		24/0/0	24/0/0	24/0/0	24/0/0	24/0/0	24/0/0	24/0/0	24/0/0	21/3/0	-

TABLE VII: F-score (%) comparisons on 6 datasets with different LERs. (See the title of Tables II for more information).

Dataset	LER	LP$_{vote}$	AMMSS$_{MC}$	AMGL$_{MC}$	AMMSS$_{KC}$	AMGL$_{KC}$	MVL-IV	DMF	iMVWL	SLIM	AMSC
BBC2	5%	91.4(1.1)•	91.3(0.9)•	91.5(1.0)•	91.2(1.1)•	91.2(1.1)•	92.4(0.8)•	92.9(0.8)•	90.6(1.5)•	90.7(1.8)•	**93.7(1.1)**
	10%	92.6(0.7)•	93.1(0.8)•	93.1(0.8)•	93.0(0.9)•	93.0(1.0)•	92.7(0.4)•	93.5(0.4)•	91.6(1.4)•	94.1(0.7)⊙	**94.5(0.6)**
	15%	93.0(0.6)•	93.9(0.5)•	94.0(0.4)•	93.7(0.6)•	93.7(0.5)•	93.1(0.2)•	93.7(0.5)•	92.1(0.9)•	94.7(0.4)⊙	**95.1(0.5)**
	20%	93.2(0.5)•	94.1(0.4)•	94.0(0.5)•	94.1(0.6)•	94.2(0.6)•	93.1(0.5)•	94.0(0.4)•	91.6(1.7)•	**95.4(0.6)**⊙	95.2(0.3)
BBC3	5%	91.9(0.8)•	93.1(0.6)•	93.1(0.5)•	81.8(1.9)•	81.6(1.9)•	91.9(0.9)•	91.2(1.1)•	90.0(0.5)•	87.5(3.3)•	**93.9(0.5)**
	10%	92.9(0.4)•	94.2(0.4)•	94.2(0.4)•	89.2(1.1)•	88.9(1.2)•	92.6(0.3)•	92.1(0.6)•	90.5(0.7)•	91.9(1.0)•	**94.6(0.4)**
	15%	93.0(0.6)•	94.4(0.4)•	94.4(0.4)•	91.3(0.5)•	91.0(0.5)•	93.0(0.4)•	92.7(0.4)•	90.8(0.9)•	93.8(1.0)•	**95.0(0.5)**
	20%	93.8(0.5)•	94.8(0.4)•	94.8(0.4)•	92.2(0.8)•	92.0(0.8)•	92.8(0.3)•	93.1(0.4)•	90.8(1.0)•	94.2(0.5)•	**95.4(0.5)**
BBC4	5%	91.1(0.5)•	92.7(0.5)•	92.7(0.5)•	68.5(3.0)•	68.9(3.0)•	93.1(0.5)•	91.1(0.7)•	86.2(2.5)•	81.1(3.8)•	**93.7(0.5)**
	10%	92.2(0.4)•	93.6(0.4)•	93.6(0.4)•	78.4(1.0)•	78.7(1.0)•	93.6(0.5)•	92.2(0.6)•	87.5(1.4)•	87.1(1.5)•	**94.5(0.3)**
	15%	92.6(0.5)•	93.9(0.4)•	93.9(0.4)•	83.2(1.4)•	83.1(1.6)•	93.5(0.3)•	92.7(0.6)•	88.2(1.5)•	91.1(1.1)•	**94.8(0.3)**
	20%	92.8(0.4)•	94.0(0.3)•	93.9(0.3)•	85.8(0.6)•	85.8(0.5)•	93.6(0.4)•	92.9(0.4)•	86.9(2.2)•	92.7(0.6)•	**94.9(0.2)**
BBCSport2	5%	86.5(3.8)•	68.2(8.5)•	73.0(7.5)•	78.6(5.6)•	80.5(5.9)•	85.6(2.5)•	92.5(1.6)•	83.8(2.6)•	77.0(3.2)•	**95.6(0.6)**
	10%	92.2(0.7)•	90.4(1.3)•	91.1(1.2)•	87.5(2.0)•	88.7(2.1)•	87.6(1.3)•	92.9(0.5)•	86.8(2.1)•	81.2(1.9)•	**96.3(0.4)**
	15%	93.1(0.7)•	93.6(0.6)•	93.9(0.6)•	92.2(0.9)•	92.7(0.7)•	87.5(1.3)•	93.1(0.6)•	86.9(2.8)•	84.0(0.9)•	**96.5(0.2)**
	20%	94.5(0.5)•	94.8(0.7)•	94.9(0.7)•	93.6(0.7)•	94.0(0.8)•	88.7(1.1)•	93.3(0.4)•	85.5(2.3)•	87.0(1.3)•	**96.9(0.4)**
BBCSport3	5%	83.8(2.4)•	77.6(5.7)•	78.5(5.2)•	51.2(2.3)•	51.5(2.1)•	87.7(0.9)•	91.1(0.9)•	73.7(5.5)•	74.1(2.2)•	**93.4(0.7)**
	10%	90.2(1.2)•	89.1(1.3)•	89.4(1.3)•	60.8(3.7)•	61.3(3.6)•	88.6(0.8)•	90.8(1.0)•	79.0(3.9)•	79.9(1.6)•	**94.4(0.9)**
	15%	91.9(1.0)•	91.9(0.9)•	91.8(0.9)•	68.5(1.1)•	68.8(1.0)•	88.8(0.5)•	91.2(0.7)•	79.0(3.1)•	82.5(1.5)•	**95.2(0.5)**
	20%	91.8(0.6)•	92.6(0.5)•	92.4(0.4)•	71.7(0.7)•	72.2(0.6)•	88.6(0.5)•	91.6(0.9)•	78.1(3.3)•	84.8(1.1)•	**95.5(0.3)**
BBCSport4	5%	82.6(2.7)•	85.2(2.8)•	85.4(2.6)•	39.5(3.6)•	40.4(3.7)•	86.7(0.5)•	91.5(0.8)•	66.6(6.5)•	72.6(1.9)•	**92.3(0.6)**
	10%	87.8(0.6)•	91.5(0.7)•	91.5(0.7)•	58.7(1.4)•	59.0(1.4)•	88.4(0.2)•	92.1(0.5)•	71.9(3.8)•	77.4(1.3)•	**93.2(0.6)**
	15%	90.1(0.7)•	92.8(0.6)•	92.8(0.5)•	67.8(0.8)•	68.0(0.6)•	88.7(0.5)•	91.9(0.5)•	72.4(3.1)•	80.2(1.2)•	**94.3(0.4)**
	20%	91.6(0.5)•	92.9(0.3)•	92.8(0.3)•	70.0(0.4)•	70.3(0.4)•	88.5(0.4)•	91.3(0.4)•	73.4(2.7)•	82.4(0.8)•	**94.8(0.3)**
win/tie/lose		24/0/0	24/0/0	24/0/0	24/0/0	24/0/0	24/0/0	24/0/0	24/0/0	21/3/0	-

the performance tends to be decreased, and on KSA and MINIST, it tends to be increased. Moreover, increasing number of anchors also makes the time cost increase significantly. Considering both effectiveness and efficiency, $m^{(v)} = 0.1n^{(v)}$ achieves satisfactory performance on these four datasets.

In our experiments, each dataset is split as labeled examples, unlabeled examples and the validation set, and the parameters are determined according to the results of the validation set.

VI. Application to Document Classification

Document classification is the process of assigning a document to a predefined set of categories based on the document content [57]. BBC and BBCSport are two news article datasets introduced by [58]. BBC consists of 2225 documents belonging to 5 classes, i.e., business (510 documents), entertainment (386 documents), politics (417 documents), sport (511 documents) and tech (401 documents). BBCSport contains 737 documents belonging to 5 classes, i.e., athletics (101 documents), cricket (124 documents), football (265 documents), rugby (147 documents) and tennis (100 documents). Document classification is more complex than other classification tasks because more time and effort are required for pre-processing data [59]. In [60], each raw document is split into 1-4 segments by merging consecutive paragraphs, and each segment with at least 200 words is assigned to at most one view. Since each document can be represented by segments in some but not

all of views, this pre-processing methodology results in six incomplete multi-view datasets[10], i.e., BBC2, BBC3, BBC4, BBCSport2, BBCSport3 and BBCSport4. The brief summary of these six datasets are listed in Table VI.

In this section, we classify the six incomplete datasets in semi-supervised scenario. Similarly, we compare AMSC with aforementioned 8 baselines. For AMSC, we set $m^{(v)} = n^{(v)}$. And other parameters are determined by the same way as experiment section. In this section, we evaluate the classification performance under different labeled example ratios (LER). The labeled example ratio varies from 5% to 20% with a step 5%. On each label example ratio, we repeat to select examples to assign labels for 10 independent times, and report the average Acc and F-score results with standard deviation (STD) in Table VI and Table VII, respectively.

From Table VI and VII, we have the following observations: (1) With the increase of labeled example ratio, the classification performance of each method tends to be better in most cases. (2) The proposed AMSC outperforms other compared methods in terms of both Acc and F-score on all datasets, indicating the effectiveness of AMSC to deal with the incomplete multi-view datasets. (3) AMMSS$_{MC}$ and AMGL$_{MC}$ achieves better performance than AMMSS$_{KC}$ and AMGL$_{KC}$ on these datasets except BBC2, indicating that data matrix completion is more suitable than graph matrix completion to recover the missing information of these news data. (4) DMF outperforms MVL-IV on BBCSport2, BBCSport3 and BBCSport4. The possible reason is that DMF can better disclose the nonlinear structure of data. (5) SLIM achieves comparable performance with AMSC on BBC2, but performs significantly worse on BBCSport2, BBCSport3 and BBCSport4. It is probably because that SLIM cannot well utilize the local information of partial similarity matrices.

VII. Conclusion

In this paper, to effectively and efficiently deal with incomplete multi-view data in semi-supervised scenario, a method named as AMSC is proposed. To utilize both complementary information and consensus information, AMSC learn view-specific label matrices and a common probability label matrix simultaneously. On each view, AMSC construct a partial graph matrix by anchor strategy, which measures the intra-view similarity between each pair rows of view-specific label matrices. And to obtain unified classification results for all unlabeled instances, extra-view similarity is defined based on common label matrix, which measures the relationships between each row of view-specific label matrices and class indicator vectors. Besides, the p-th root integration strategy is adopted to incorporate the losses of multiple views, which characterizes the contributions of different views. By analyzing the connection between the p-th root strategy and exponential decay strategy, an efficient and iterative algorithm with proved convergence is developed for optimization. The proposed method is evaluated on nine real-world datasets and applied to document classification, and its performance is remarkable in most cases. In the future, we will study how to extend AMSC to solve the incomplete multi-view clustering task or deep incomplete multi-view semi-supervised classification method.

[10]http://mlg.ucd.ie/datasets/segment.html.

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